MICROLOCAL SINGULAR SUPPORT AND SAITO'S CONJECTURE ON CHARACTERISTIC CLASSES

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Abstract. Let X be a smooth scheme over a perfect field of characteristic $p > 0$. We prove that the microlocal singular support defined by Saito is contained in the singular support defined by Beilinson for every constructible complex of finite Tor-dimension with finite coefficients prime to p. The same assertion holds for constructible complexes with coefficients in a finite extension of $\mathbf{Q}_\ell$. The proof is based on a proper deformation of the collision locus in the double incidence correspondence and the specialization of the Radon unit and counit. As an application, for every closed immersion $Z \hookrightarrow X$ of smooth schemes, we prove

$$\operatorname{supp} \mu_Z(F) \subseteq \mathrm{SS}(F) \cap T^*_Z X.$$

For perverse E-sheaves, we prove that the two singular supports are equal. We also show that the microlocal image of the identity endomorphism has support equal to this common singular support, which is an étale analogue of a theorem of Kashiwara–Schapira.

We further compare the microlocal characteristic class with the cycle class of the characteristic cycle. Combining this comparison with the two zero-section formulas, we confirm Saito's conjecture, namely that the cohomological characteristic classes defined by Abbes and Saito can be computed in terms of the characteristic cycles.

We also prove a Künneth formula for microlocal Hom and the multiplicativity of microlocal singular support and microlocal characteristic class under external products.

Contents

1. Introduction

1.1. Let k be a perfect field of characteristic $p > 0$, and let X be a smooth k-scheme. The singular support of a constructible sheaf records the cotangent directions in which the sheaf fails to be locally acyclic. In positive characteristic, Beilinson defines this invariant by means of the Radon transform [Bei16]. More precisely, if Λ is a finite local ring of residue characteristic $\ell \neq p$ and $F \in D^b_{\mathrm{ctf}}(X, \Lambda)$,

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he associates with F a closed conical subset

$$\mathrm{SS}(F) \subset T^*X.$$

It is characterized by universal local acyclicity of test pairs. If X is connected, every irreducible component of $\mathrm{SS}(F)$ has dimension $\dim X$.

Saito gives a different construction, starting from the identity correspondence of F [Sai26]. Let $\Delta : X \hookrightarrow X \times X$ be the diagonal and put

$$\mathscr{H}_F = R\mathcal{H}om(\mathrm{pr}_1^* F, \mathrm{pr}_2^! F). \tag{1.1.1}$$

After replacing Λ by a finite faithfully flat extension, we fix a nontrivial character $\psi : \mathbf{F}_p \to \Lambda^\times$. Verdier specialization of $\mathscr{H}_F$ along Δ is a monodromic complex on the normal bundle $N_{\Delta/(X\times X)} \simeq TX$. Saito defines

$$\mu\mathcal{H}om(F, F) = \mathfrak{F}_\psi \nu_\Delta(\mathscr{H}_F), \qquad \mathrm{SS}_\mu(F) = \mathrm{pr}_{T^*X} \operatorname{supp} \mu\mathcal{H}om(F, F). \tag{1.1.2}$$

Here the geometric generic-point factor in Verdier specialization is suppressed. We call $\mathrm{SS}_\mu(F)$ the microlocal singular support of F.

The two supports are defined in apparently different ways. Beilinson's support is detected by functions and local acyclicity, whereas Saito's support is obtained by specialization and Fourier transform. Saito asks whether these constructions are related as follows.

Question 1.2 (Saito, [Sai26])**.** For every constructible complex F on a smooth k-scheme, does one have $\mathrm{SS}_\mu(F) \subset \mathrm{SS}(F)$?

It is also natural to ask when the inclusion is an equality and how the corresponding characteristic classes are related. The purpose of this paper is to answer these questions.

1.3. Our first main result gives an affirmative answer to Question 1.2.

Theorem 1.4 (Theorem 5.4)**.** *For every smooth k-scheme X and every $F \in D^b_{\mathrm{ctf}}(X, \Lambda)$,*

$$\mathrm{SS}_\mu(F) \subset \mathrm{SS}(F). \tag{1.4.1}$$

The assertion also holds for constructible complexes with coefficients in a finite extension $E/\mathbf{Q}_\ell$, interpreted on the pro-étale site.

We prove the finite-coefficient assertion first. The result for E-coefficients is formulated in the constructible derived category on the pro-étale site. We use Barrett's E-adic Radon description of singular support, and pass from finite reductions to integral and rational coefficients by derived completeness [BS15, Bar24]. Since a finite faithfully flat extension of the coefficient ring does not change a conical support, the auxiliary choice of ψ causes no descent problem. In particular, Barrett's dimension theorem for singular support gives

$$\dim \mathrm{SS}_\mu(F) \leq \dim X. \tag{1.4.2}$$

The theorem also answers a second question of Saito. Let $i : Z \hookrightarrow X$ be a closed immersion of smooth k-schemes, put $N = N_{Z/X}$, and define $\mu_Z(F) = \mathfrak{F}_{\psi, N/Z}(\nu_Z F)$ on $N^\vee = T^*_Z X$. Saito asks whether its support is contained in $\mathrm{SS}(F) \cap T^*_Z X$ [Sai26, Question 2.5.3]. The deformation of the pair groupoid $X \times X \rightrightarrows X$ along its diagonal acts on the deformation of X along Z. After applying nearby cycles and Fourier transform, this action becomes

$$(q^\vee)^* \mu\mathcal{H}om(F, F) \otimes \mu_Z(F) \longrightarrow \mu_Z(F),$$

where $q : T_X|_Z \twoheadrightarrow N$ is the quotient map. If a stalk of $\mu_Z(F)$ is nonzero, the unit of this action forces the corresponding stalk of $\mu\mathcal{H}om(F, F)$ to be nonzero. Hence

$$\operatorname{supp} \mu_Z(F) \subset \mathrm{SS}_\mu(F) \cap T^*_Z X \subset \mathrm{SS}(F) \cap T^*_Z X. \tag{1.4.3}$$

The action and its unit are constructed in Subsection 5.6.

1.5. We explain the main idea of the proof. After a projective reduction, we may work on a projective space P. Let $P^\vee$ be the dual projective space and let

$$Q \subset P \times P^\vee, \qquad p : Q \longrightarrow P,$$

be the incidence correspondence and its first projection. Beilinson's Radon theorem identifies the projectivization of $\mathrm{SS}(F)$ with the non-ULA locus of the Radon transform. Thus it is enough to show that $\mu\mathcal{H}om(F, F)$ vanishes on the cotangent lines parametrized by the ULA locus.

A direct comparison encounters a properness problem. Indeed, deformation of the diagonal $\Delta_Q \subset Q \times Q$ induces the nonproper map $dp : TQ \to TP$ on the special fiber. Proper base change therefore cannot be applied to the Radon trace. The point of our construction is to enlarge the center of deformation.

We use the full collision locus

$$C = (p \times p)^{-1}(\Delta_P) = Q \times_P Q. \tag{1.5.1}$$

It remembers the common point of P together with both moving hyperplanes. Since p is flat, deformation to C is the base change of deformation to Δ_P. Since p is projective, the resulting morphism of deformation spaces is proper. This simple change of center allows us to specialize the unit and counit of Radon inversion.

Let G be the incidence pullback of the Radon transform of F. Proper duality and proper base change give a retraction

$$\nu_{\Delta_P}(\mathscr{H}_F) \longrightarrow R\varrho_* \nu_C(\mathscr{H}_G) \longrightarrow \nu_{\Delta_P}(\mathscr{H}_F). \tag{1.5.2}$$

The normal bundle of C is $C \times_P TP$. On this bundle, localization with respect to the Radon non-ULA locus separates the specialized kernel into two terms. The open term is constant in the tangent direction, and its Fourier transform is therefore supported on the zero section. The closed term is strongly equivariant under the translation subbundles determined by the two hyperplanes. Fourier orthogonality then restricts its nonzero support to the tautological cotangent lines above the Radon non-ULA locus. The retraction (1.5.2) transfers this support estimate to $\nu_{\Delta_P}(\mathscr{H}_F)$. Beilinson's Radon criterion proves (1.4.1). This argument uses neither a boundary transversality assumption nor a tame-monodromy assumption.

1.6. We next consider the reverse inclusion. For a perverse sheaf with characteristic-zero coefficients, every irreducible component of the singular support can be detected by a suitable pencil. Combining this fact with Theorem 1.4, we obtain an equality.

Theorem 1.7 (Theorem 6.7)**.** *Let E be a finite extension of $\mathbf{Q}_\ell$. For every perverse E-sheaf $\mathcal{P}$ on a smooth k-scheme,*

$$\mathrm{SS}_\mu(\mathcal{P}) = \mathrm{SS}(\mathcal{P}). \tag{1.7.1}$$

Here is the contradiction argument. Suppose that an irreducible component of $\mathrm{SS}(\mathcal{P})$ is not contained in $\mathrm{SS}_\mu(\mathcal{P})$. After a Veronese embedding, Barrett's E-adic Radon theorem detects this component by a generically radicial component of the Radon discriminant. A general line through a suitable closed point defines a pencil with a unique non-ULA point and nonzero vanishing cycles. On the other hand, the same pencil is non-characteristic for $\mathrm{SS}_\mu(\mathcal{P})$. Its microlocal intersection number, and hence its Artin conductor, is zero. Saito's Milnor formula identifies this conductor with the nonzero total dimension of the unique vanishing-cycle complex, a contradiction [Sai17, Bar24]. Proper base change in the pro-étale category is obtained by reduction modulo the powers of a uniformizer and derived completeness.

For finite coefficients, the non-acyclicity class of Yang–Zhao provides a parallel obstruction. Its cohomological Milnor formula detects the missing component whenever the resulting local class is nonzero; see Corollary 6.3 [YZ25].

1.8. The dimension estimate (1.4.2) has a further consequence. It verifies the hypothesis in Saito's comparison theorem and gives

$$\mathrm{CC}_\mu(F) = \mathrm{cl}(\mathrm{CC}(F)) \tag{1.8.1}$$

for finite and E-adic coefficients. Thus the microlocal characteristic class is the cycle class of the characteristic cycle. For E-coefficients, we choose a torsion-free perverse integral model. Perverse exactness of vanishing cycles and compatibility of wild inertia with change of coefficients preserve total dimension. Derived completeness then passes the finite-level identity to integral and rational coefficients.

The identity endomorphism of F defines, by specialization and Fourier transform, a canonical section

$$s_F : \Lambda_{T^*X} \longrightarrow \mu\mathcal{H}om(F, F),$$

with Λ replaced by E in the pro-étale setting. The class $\mathrm{CC}_\mu(F)$ is the image of this section under the microlocal evaluation morphism. If $\mathcal{P}$ is a perverse E-sheaf, the positivity of the coefficients of $\mathrm{CC}(\mathcal{P})$ and (1.8.1) imply

$$\mathrm{supp}(s_{\mathcal{P}}) = \mathrm{supp}\, \mu\mathcal{H}om(\mathcal{P}, \mathcal{P}) = \mathrm{SS}(\mathcal{P}). \tag{1.8.2}$$

This is the ètale counterpart of [KS90, Corollary 6.1.3]. For finite coefficients, the trace of the identity may vanish modulo ℓ; therefore the trace class alone does not prove the first equality in (1.8.2).

For finite coefficients, pullback by the zero section identifies the left side of (1.8.1) with the cohomological characteristic class. The same pullback on the right is the cycle class of the zero-dimensional part of the characteristic cycle. We therefore obtain Saito's conjecture on cohomological characteristic classes, which we now recall.

Conjecture 1.9 (Saito, [Sai17, Conjecture 6.8])**.** *Let Z be a finite-type k-scheme admitting a closed immersion into a smooth finite-type k-scheme. For every $F \in D^b_{\mathrm{ctf}}(Z, \Lambda)$, one has*

$$C_{Z/k}(F) = \mathrm{cl}_Z(cc_{Z,0}(F)) \quad \text{in } H^0(Z, K_{Z/k}). \tag{1.9.1}$$

We prove the following theorem.

Theorem 1.10 (Theorem 7.9)**.** *For the finite coefficient ring Λ fixed above, Conjecture 1.9 holds.*

Theorems 1.7 and 1.10 are two different consequences of Theorem 1.4. The first uses a detecting pencil, the microlocal conductor formula, and positivity of total dimension. The second uses the dimension bound and the two zero-section formulas. The non-acyclicity class also yields localization, fibration, and proper push-forward formulas for the cohomological characteristic class; we collect these applications in Section 6.

Finally, the construction is compatible with external products. For smooth k-schemes X, Y and constructible complexes F, G, we prove a canonical isomorphism

$$\mu\mathcal{H}om(F \boxtimes G, F \boxtimes G) \simeq \mu\mathcal{H}om(F, F) \boxtimes \mu\mathcal{H}om(G, G).$$

Consequently,

$$\mathrm{SS}_\mu(F \boxtimes G) = \mathrm{SS}_\mu(F) \times \mathrm{SS}_\mu(G), \qquad \mathrm{CC}_\mu(F \boxtimes G) = \mathrm{CC}_\mu(F) \boxtimes \mathrm{CC}_\mu(G).$$

1.11. The paper is organized as follows. Section 2 fixes our conventions and recalls Beilinson's Radon criterion. In Section 3, we construct the proper collision deformation and specialize the Radon retraction. In Section 4, we prove the Fourier support estimates on the collision normal bundle. These results are combined in Section 5 to prove Theorem 1.4 and the specialization estimate (1.4.3). In Section 6, we study the non-acyclicity class and prove Theorem 1.7. Section 7 constructs the microlocal identity section, proves (1.8.2) and (1.8.1), and deduces Saito's conjecture on cohomological characteristic classes. In Section 8, we establish the microlocal Künneth formula and its consequences.

Notation and Conventions.

(1) All schemes are separated and of finite type over the fixed perfect field k of characteristic $p > 0$, unless otherwise stated.

(2) The coefficient ring Λ is a finite local ring killed by a power of a prime $\ell \neq p$. The notation $D^b_{\mathrm{ctf}}(X, \Lambda)$ denotes the derived category of constructible complexes of finite Tor dimension. If $E/\mathbf{Q}_\ell$ is finite with ring of integers $\mathcal{O}_E$, then $D_{\mathrm{cons}}(X_{\text{proét}}, \mathcal{O}_E)$ and $D_{\mathrm{cons}}(X_{\text{proét}}, E)$ denote the constructible pro-étale categories of Bhatt–Scholze, Hemo–Richarz–Scholbach, and Hansen–Scholze [BS15, HRS23, HS23]. They satisfy

$$D_{\mathrm{cons}}(X_{\text{proét}}, \mathcal{O}_E) \simeq \varprojlim_m D^b_{\mathrm{ctf}}(X, \mathcal{O}_E/\pi^m), \qquad D_{\mathrm{cons}}(X_{\text{proét}}, E) \simeq D_{\mathrm{cons}}(X_{\text{proét}}, \mathcal{O}_E)[1/\pi]$$

up to the standard idempotent completion. All E-adic operations below are taken in this intrinsic category. Integral models and their reductions are used to verify identities, not to define the rational object or its singular support.

(3) After a finite faithfully flat extension of Λ, we fix a nontrivial character $\psi : \mathbf{F}_p \to \Lambda^\times$. The extension is descended at the end of the argument. Changing ψ acts on the cotangent fibers by a nonzero scalar and therefore does not change any conical support.

(4) For a separated morphism $a : X \to S$, write $K_{X/S} = Ra^!\Lambda$ and $\mathbb{D}_{X/S}(-) = R\mathcal{H}om(-, K_{X/S})$. We write $\mathbb{D}_X$ when the base is $\operatorname{Spec} k$.

(5) Verdier specialization is taken over the geometric generic point of the henselization of $\mathbf{A}^1$ at the origin. This factor is suppressed when it does not affect an argument. Shifts and Tate twists in Fourier normalization do not affect support. Nevertheless, characteristic classes are always formed with Saito's unshifted Fourier transform fixed in (2.1.3).

(6) For a closed conical subset of a vector bundle, supp denotes its set-theoretic support. All pullbacks and direct images are derived; the symbols R and L are retained when they clarify the direction of a comparison.

2. Microlocalization and the Radon criterion

2.1. We first recall the deformation to the normal cone, Verdier specialization, and the Fourier–Deligne transform. Let $i : Z \hookrightarrow Y$ be a closed immersion defined by a quasi-coherent ideal $I \subset \mathcal{O}_Y$, and let τ denote the coordinate of $\mathbf{A}^1$. We put

$$\mathscr{D}_Z(Y) = \operatorname{Spec}_Y \mathcal{R}_I^{\mathrm{ext}}, \qquad \mathcal{R}_I^{\mathrm{ext}} = \mathcal{O}_Y[\tau, I\tau^{-1}] = \bigoplus_{m \in \mathbf{Z}} I^m \tau^{-m} \subset \mathcal{O}_Y[\tau, \tau^{-1}], \tag{2.1.1}$$

where $I^m = \mathcal{O}_Y$ for $m \leq 0$. Equivalently, $\mathscr{D}_Z(Y)$ is the complement of the strict transform of $Y \times \{0\}$ in $\mathrm{Bl}_{Z \times \{0\}}(Y \times \mathbf{A}^1)$. The element τ induces a morphism

$$\pi_{\mathscr{D}} : \mathscr{D}_Z(Y) \longrightarrow \mathbf{A}^1.$$

Its restriction over $\mathbf{G}_m$ is canonically identified with $Y \times \mathbf{G}_m$, and its fiber over 0 is the normal cone

$$C_{Z/Y} = \operatorname{Spec}_Z \left(\bigoplus_{m \geq 0} I^m / I^{m+1} \right).$$

If i is a regular immersion, this cone is the normal bundle $N_{Z/Y} = \operatorname{Spec}_Z \operatorname{Sym}(I/I^2)$. All the immersions considered below are regular. Let $j : Y \times \mathbf{G}_m \hookrightarrow \mathscr{D}_Z(Y)$ and $i_0 : N_{Z/Y} \hookrightarrow \mathscr{D}_Z(Y)$ denote the generic and special immersions, respectively. For $K \in D^b_c(Y, \Lambda)$, we define its Verdier specialization along Z by

$$\nu_Z(K) = \Psi(K \boxtimes \Lambda_{\mathbf{G}_m}) \qquad \text{in} \qquad D^b_{\mathrm{ctf}}(C_{Z/Y} \times_k \eta_0, \Lambda), \tag{2.1.2}$$

where η_0 is the geometric generic point of the henselization of $\mathbf{A}^1$ at 0.

Let $e : V \to S$ be a vector bundle of rank r and let $V^\vee$ be its dual. We denote by q_1, q_2 the projections of $V \times_S V^\vee$ and by $\langle \cdot, \cdot \rangle$ the canonical pairing. We define the unshifted Fourier–Deligne transform by

$$\mathfrak{F}_{\psi,V/S}(K) = Rq_{2!}\big(q_1^* K \otimes \mathcal{L}_\psi(\langle v, \xi \rangle)\big). \tag{2.1.3}$$

This is the normalization used by Saito. The normalized transform is $\mathfrak{F}_{\psi,V/S}[r]$. It has the same support as $\mathfrak{F}_{\psi,V/S}$, but it is not used in the definition of CC_μ. This convention is needed for the degree-zero characteristic-class formulas; Tate twists do not affect the support statements.

2.2. Let X be a smooth scheme X over k. Put $Z = \Delta_X \subset X \times X$. We identify $N_{Z/(X\times X)}$ with TX. For an object $F \in D^b_{ctf}(X, \Lambda)$, we define:

$$\mathscr{H}_F = R\mathcal{H}om(\mathrm{pr}_1^* F, \mathrm{pr}_2^! F), \tag{2.2.1}$$

$$\mu\mathcal{H}om(F, F) = \mathfrak{F}_\psi \nu_\Delta(\mathscr{H}_F), \tag{2.2.2}$$

$$\mathrm{SS}_\mu(F) = \mathrm{pr}_{T^*X} \operatorname{supp} \mu\mathcal{H}om(F, F). \tag{2.2.3}$$

2.3. We next recall Beilinson's Radon criterion and reduce theorem 1.4 to a vanishing statement on projective space. Let V be a vector space of dimension $n+1$, where $n \geq 1$, and put

$$P = \mathbf{P}(V), \qquad P^\vee = \mathbf{P}(V^\vee).$$

Consider the incidence correspondence

$$Q = \{(x, H) \in P \times P^\vee \mid x \in H\}, \qquad p : Q \to P, \qquad h : Q \to P^\vee \tag{2.3.1}$$

and put $d = n - 1$. We use the normalized Radon transform

$$\mathbf{R}(F) = Rh_* p^* F[d]. \tag{2.3.2}$$

For a morphism $a : Y \to S$ and $K \in D^b_c(Y, \Lambda)$, let $E_a(K)$ denote the smallest closed subset of Y such that a is universally locally acyclic relative to K over its complement.

For $F \in D^b_{\mathrm{ctf}}(P, \Lambda)$, we put

$$E_F^{\mathrm{Rad}} = E_p(h^* \mathbf{R}(F)) \subset Q. \tag{2.3.3}$$

By Beilinson's Radon theorem [Bei16, Theorem 3.2], the Legendre isomorphism

$$Q \xrightarrow{\sim} \mathbf{P}(T^*P), \qquad (x, H) \longmapsto [T_H^* P|_x] \tag{2.3.4}$$

identifies E_F^{Rad} with $\mathbf{P}(\mathrm{SS}(F))$. Let

$$\rho : T^*P \setminus T_P^* P \longrightarrow \mathbf{P}(T^*P) \simeq Q \tag{2.3.5}$$

be the quotient by scalar dilation. It follows that the nonzero part of theorem 1.4 for P is equivalent to

$$\mathrm{SS}_\mu(F) \setminus T_P^* P \subset \rho^{-1}(E_F^{\mathrm{Rad}}). \tag{2.3.6}$$

Proposition 2.4 (Projective reduction)**.** *If* (2.3.6) *holds for every constructible complex on projective space, then theorem 1.4 holds for every smooth finite-type k-scheme.*

Proof. We first treat the zero-section part. If F vanishes on an open neighborhood U of $x \in P$, then $\mathscr{H}_F$ vanishes on a neighborhood of Δ_U. Its specialization and its Fourier transform therefore vanish over U. Thus the base of $\mathrm{SS}_\mu(F)$ is contained in $\operatorname{supp} F$. On the other hand, the base of $\mathrm{SS}(F)$ is $\operatorname{supp} F$ by [Bei16, Section 2.3(iii)], and the zero covector over every point of $\operatorname{supp} F$ belongs to the closed conical subset $\mathrm{SS}(F)$. Hence (2.3.6) implies the assertion for P, including the zero-section part.

Let now X be a smooth finite-type k-scheme. Since the assertion is local on X, we may assume that X is affine and choose a closed immersion $i : X \hookrightarrow \mathbf{A}^N = U$. Let $j : U \hookrightarrow \mathbf{P}^N$ be the standard open immersion and put $\overline{F} = j_! i_* F$. Applying the projective case to $\overline{F}$ and restricting to U, we obtain the assertion for i_*F. Indeed, open restriction is compatible with both singular supports by [Bei16, Section 2.3(i)] and [Sai26, Lemma 2.4.1]. For the closed immersion i, we have

$$\mathrm{SS}_\mu(i_*F) = i_\circ \mathrm{SS}_\mu(F), \qquad \mathrm{SS}(i_*F) = i_\circ \mathrm{SS}(F)$$

by [Sai26, Lemma 2.3.1] and [Bei16, Section 2.5(ii)]. The operation $i_\circ$ is induced by $T^*U|_X \to T^*X$ and reflects inclusions. More precisely, if $q : T^*U|_X \twoheadrightarrow T^*X$ is the quotient map, then $i_\circ C = q^{-1}(C)$ as a subset of $T^*U|_X \subset T^*U$. The result for i_*F therefore implies the result for F. If the ground field is finite, the closed-immersion equality for SS follows after a finite extension of the ground field as in the last paragraph of [Bei16, Section 2.5(ii)]. Such an extension does not change the singular support. This proves the proposition. ☐

We fix an open subset $W \subset Q \setminus E_F^{\mathrm{Rad}}$. By the definition of E_F^{Rad}, we have

$$h^*\mathbf{R}(F)|_W \quad \text{is ULA over } P. \tag{2.4.1}$$

In view of (2.3.6), it remains to prove that $\mu\mathcal{H}om(F,F)$ vanishes on the nonzero cotangent lines parametrized by W.

3. The proper collision deformation and the Radon retraction

3.1. We first construct a proper comparison between the Radon correspondence and Verdier specialization. Consider the notation in 2.3. We have a cartesian diagram

$$\begin{tikzcd} C = Q \times_P Q \arrow[r, hook] \arrow[d] & Q \times Q \arrow[d, "p\times p"] \\ \Delta_P \arrow[r, hook] & P \times P. \end{tikzcd} \tag{3.1.1}$$

The morphism $p \times p$ induces a morphism between the deformations to the normal cones

$$\Pi : \mathscr{D}_C(Q \times Q) \longrightarrow \mathscr{D}_{\Delta_P}(P \times P). \tag{3.1.2}$$

More explicitly, the source and the target of (3.1.2) are

$$\mathscr{D}_C(Q\times Q) = \mathrm{Spec}_{Q\times Q}\, \mathcal{O}_{Q\times Q}[\tau, I_C\tau^{-1}], \qquad \mathscr{D}_{\Delta_P}(P\times P) = \mathrm{Spec}_{P\times P}\, \mathcal{O}_{P\times P}[\tau, I_{\Delta_P}\tau^{-1}].$$

The map Π is induced by $p \times p$ and the equality $I_C = I_{\Delta_P}\mathcal{O}_{Q\times Q}$.

The incidence projection $p : Q \to P$ is a $\mathbf{P}^{n-1}$-bundle. It is therefore smooth and projective. Thus we have the following result (see also [Sai26, (1.50)]).

Lemma 3.2 (Proper moving-collision deformation)**.** *There is a cartesian square*

$$\begin{tikzcd} \mathscr{D}_C(Q\times Q) \arrow[r, "\Pi"] \arrow[d] & \mathscr{D}_{\Delta_P}(P\times P) \arrow[d] \\ Q\times Q \arrow[r, "p\times p"'] & P\times P. \end{tikzcd} \tag{3.2.1}$$

Consequently Π *is proper. On the special fiber it is*

$$\varrho : N_{C/(Q\times Q)} \xrightarrow{\sim} C \times_P TP \longrightarrow TP, \qquad (y, H_1, H_2, v) \longmapsto (y, v). \tag{3.2.2}$$

Proof. We put

$$f = p\times p : Q\times Q \longrightarrow P\times P, \qquad I = I_{\Delta_P}, \qquad J = I_C.$$

Since $p : Q \to P$ is a $\mathbf{P}^{n-1}$-bundle, it is smooth, flat, and projective. The same assertions hold for f. By the definition $C = (Q\times Q)\times_{P\times P}\Delta_P$, we have, scheme-theoretically,

$$J = I\mathcal{O}_{Q\times Q}. \tag{3.2.3}$$

We prove the cartesian assertion by comparing the graded pieces of the extended Rees algebras. For every $m \geq 0$, the flatness of f implies that the natural map

$$I^m \otimes_{\mathcal{O}_{P\times P}} \mathcal{O}_{Q\times Q} \longrightarrow \mathcal{O}_{Q\times Q}$$

is injective. Its image is $I^m\mathcal{O}_{Q\times Q}$, and

$$I^m\mathcal{O}_{Q\times Q} = (I\mathcal{O}_{Q\times Q})^m = J^m.$$

Consequently

$$I^m \otimes_{\mathcal{O}_{P\times P}} \mathcal{O}_{Q\times Q} \xrightarrow{\sim} J^m. \tag{3.2.4}$$

For $m \leq 0$, the same assertion holds by the convention that both ideal powers are the corresponding structure sheaves. Since tensor products commute with direct sums, (3.2.4) gives an isomorphism of extended Rees algebras

$$\begin{aligned} \mathcal{R}_I^{\text{ext}} \otimes_{\mathcal{O}_{P\times P}} \mathcal{O}_{Q\times Q} &= \left(\bigoplus_{m\in\mathbf{Z}} I^m\tau^{-m}\right) \otimes_{\mathcal{O}_{P\times P}} \mathcal{O}_{Q\times Q} \\ &\xrightarrow{\sim} \bigoplus_{m\in\mathbf{Z}} J^m\tau^{-m} = \mathcal{R}_J^{\text{ext}}. \end{aligned} \tag{3.2.5}$$

Taking relative spectra, we obtain a canonical isomorphism over $\mathbf{A}^1$

$$\mathscr{D}_C(Q\times Q) \simeq (Q\times Q)\times_{P\times P}\mathscr{D}_{\Delta_P}(P\times P).$$

This proves that (3.2.1) is cartesian. Since Π is a base change of the projective morphism f, it is projective and, in particular, proper.

It remains to identify the special fiber. Tensoring the exact sequence

$$0 \longrightarrow I^{m+1} \longrightarrow I^m \longrightarrow I^m/I^{m+1} \longrightarrow 0$$

with the flat $\mathcal{O}_{P\times P}$-algebra $\mathcal{O}_{Q\times Q}$, and then applying (3.2.4), we obtain

$$(I^m/I^{m+1}) \otimes_{\mathcal{O}_{P\times P}} \mathcal{O}_{Q\times Q} \xrightarrow{\sim} J^m/J^{m+1}.$$

It follows that the associated graded algebras, and hence the normal cones, satisfy

$$C_{C/(Q\times Q)} \simeq C \times_{\Delta_P} C_{\Delta_P/(P\times P)}.$$

Since P is smooth, $\Delta_P \hookrightarrow P\times P$ is a regular immersion. Its flat pullback $C \hookrightarrow Q\times Q$ is also regular. Consequently the two normal cones are normal bundles. By the canonical difference-of-tangent-vectors identification, we have

$$N_{C/(Q\times Q)} \simeq C \times_{\Delta_P} N_{\Delta_P/(P\times P)} \simeq C \times_P TP.$$

Under this identification, the special-fiber map induced by Π is the projection

$$C \times_P TP \longrightarrow TP, \qquad (y, H_1, H_2, v) \longmapsto (y, v),$$

which is the morphism (3.2.2). □

Remark 3.3. The preceding construction requires the full center C. Indeed, the special fiber of the morphism obtained from the smaller center Δ_Q, $\mathscr{D}_{\Delta_Q}(Q\times Q) \to \mathscr{D}_{\Delta_P}(P\times P)$ is $dp : TQ \to TP$. Its fibers contain the positive-dimensional affine spaces $T_{Q/P}$, so this morphism is not proper. The points (q_1, q_2) satisfying $q_1 \neq q_2$ and $p(q_1) = p(q_2)$ are precisely the omitted collision terms; the center C retains these points.

3.4. We next compare the endomorphism kernels before and after proper specialization. For $G_1, G_2 \in D^b_{\mathrm{ctf}}(Q,\Lambda)$, put $H_i = Rp_*G_i$ and define

$$\mathscr{H}_Q(G_1,G_2) = R\mathcal{H}om(\mathrm{pr}_1^* G_1, \mathrm{pr}_2^! G_2), \tag{3.4.1}$$

$$\mathscr{H}_P(H_1,H_2) = R\mathcal{H}om(\mathrm{pr}_1^* H_1, \mathrm{pr}_2^! H_2). \tag{3.4.2}$$

These kernels are contravariant in the first variable and covariant in the second. If $G_1 = G_2 = G$ and $H = Rp_*G$, we write

$$\mathscr{H}_G = \mathscr{H}_Q(G,G), \qquad \mathscr{H}_H = \mathscr{H}_P(H,H).$$

Proposition 3.5 (Bifunctorial quadratic proper specialization)**.** *For every pair (G_1,G_2) there are canonical isomorphisms*

$$\alpha_{G_1,G_2}: \quad R(p\times p)_*\mathscr{H}_Q(G_1,G_2) \xrightarrow{\sim} \mathscr{H}_P(H_1,H_2), \tag{3.5.1}$$

$$\beta_{G_1,G_2}: \quad R\varrho_*\nu_C\mathscr{H}_Q(G_1,G_2) \xrightarrow{\sim} \nu_{\Delta_P}\mathscr{H}_P(H_1,H_2). \tag{3.5.2}$$

Here and below we suppress the factor η_0; thus $R\varrho_*$ in (3.5.2) means $R(\varrho\times\mathrm{id}_{\eta_0})_*$. The isomorphisms are binatural: contravariant in G_1 and covariant in G_2. They preserve the composition of cohomological correspondences and are compatible with Verdier duality. In particular, setting $G_1 = G_2 = G$ gives

$$R(p\times p)_*\mathscr{H}_G \simeq \mathscr{H}_H, \qquad R\varrho_*\nu_C(\mathscr{H}_G) \simeq \nu_{\Delta_P}(\mathscr{H}_H).$$

Proof. We first prove (3.5.1). Put $K_{Q/k} = Ra_Q^!\Lambda$ and $\mathbb{D}_Q(-) = R\mathcal{H}om(-,K_{Q/k})$, and use analogous notation on P. Since Q and P are smooth over k, smooth purity for the projection $\mathrm{pr}_2 : Q\times Q\to Q$ gives an isomorphism

$$\mathrm{pr}_2^! G_2 \simeq \mathrm{pr}_1^* K_{Q/k}\otimes \mathrm{pr}_2^* G_2. \tag{3.5.3}$$

The complex G_1 has finite Tor dimension and is therefore dualizable. Consequently we have canonical isomorphisms

$$\begin{aligned}\mathscr{H}_Q(G_1,G_2) &\simeq R\mathcal{H}om\big(\mathrm{pr}_1^* G_1, \mathrm{pr}_1^* K_{Q/k}\otimes \mathrm{pr}_2^* G_2\big)\\ &\simeq \mathrm{pr}_1^*\mathbb{D}_Q G_1\otimes \mathrm{pr}_2^* G_2 = \mathbb{D}_Q G_1\boxtimes G_2.\end{aligned} \tag{3.5.4}$$

Applying the same argument on P, we obtain

$$\mathscr{H}_P(H_1,H_2) \simeq \mathbb{D}_P H_1\boxtimes H_2. \tag{3.5.5}$$

By the proper Künneth formula and Grothendieck duality for p, we have

$$\begin{aligned}R(p\times p)_*\mathscr{H}_Q(G_1,G_2) &\simeq Rp_*\mathbb{D}_Q G_1\boxtimes Rp_*G_2\\ &\simeq \mathbb{D}_P(Rp_*G_1)\boxtimes Rp_*G_2 \simeq \mathscr{H}_P(H_1,H_2).\end{aligned} \tag{3.5.6}$$

Each isomorphism in (3.5.6) is natural contravariantly in G_1 and covariantly in G_2. This proves (3.5.1) and its binaturality.

We next prove (3.5.2). Let

$$\mathscr{T} = \operatorname{Spec}\mathcal{O}^h_{\mathbf{A}^1,0}$$

be the henselian trait at the origin, with geometric generic point η_0 and closed point s. We base change the two deformation spaces and Π to $\mathscr{T}$. By lemma 3.2, the resulting morphism is proper over $\mathscr{T}$, and its generic and special fibers are

$$\Pi_{\eta_0} = (p\times p)\times \mathrm{id}_{\eta_0}, \qquad \Pi_s = \varrho.$$

For a proper morphism over a trait, the two transformations (3.2) and (3.3) of [LZ22, Construction 3.2] are inverse to each other. Applying them to Π, for every constructible complex K on the generic fiber we obtain the canonical proper-nearby-cycle isomorphism

$$R(\Pi_s)_*\Psi_{\mathscr{D}_C}(K) \xrightarrow{\sim} \Psi_{\mathscr{D}_{\Delta_P}} R(\Pi_{\eta_0})_*K. \tag{3.5.7}$$

We apply (3.5.7) to $K = \mathscr{H}_Q(G_1, G_2) \boxtimes \Lambda_{\eta_0}$. The left side of (3.5.7) is $R\varrho_* \nu_C \mathscr{H}_Q(G_1, G_2)$, while (3.5.1) identifies its right side with $\nu_{\Delta_P} \mathscr{H}_P(H_1, H_2)$. This proves (3.5.2).

It remains to prove the two compatibility assertions. Before specialization, (3.5.6) is the comparison of mapping objects induced by proper pushforward in the 2-category of cohomological correspondences. By the uniqueness assertion in [LZ22, Construction 2.10], proper pushforward is compatible with horizontal and vertical composition. By [LZ22, Construction 3.3], nearby cycles form a symmetric monoidal functor, and [LZ22, Proposition 3.12] identifies specialization before and after proper pushforward. Applying this commutative square to each factor of a composite proves the compatibility with composition.

By proper Grothendieck duality, (3.5.6) is also compatible with Verdier duality. If σ_Q and σ_P denote the involutions exchanging the two factors, we have

$$\mathbb{D}_{Q\times Q} \mathscr{H}_Q(G_1, G_2) \simeq \sigma_Q^* \mathscr{H}_Q(G_2, G_1), \qquad \mathbb{D}_{P\times P} \mathscr{H}_P(H_1, H_2) \simeq \sigma_P^* \mathscr{H}_P(H_2, H_1).$$

By [LZ22, Corollary 3.8], nearby cycles commute with Verdier duality. Hence (3.5.2) is compatible with these identifications. This proves the proposition. □

3.6. We now construct the retraction on the specialized endomorphism kernels. Put $d = n - 1$ and define the reverse normalized Radon transform by

$$\mathbf{R}^\vee(K) = Rp_* h^* K[d], \qquad \mathsf{S} = \mathbf{R}^\vee(d). \tag{3.6.1}$$

Thus $\mathbf{R} = Rh_* p^*[d]$. By Beilinson's Radon adjunction [Bei16, Section 1.6.1], S is both a left and a right adjoint of $\mathbf{R}$. We apply proposition 3.5 to

$$G = h^* \mathbf{R}(F)[d](d), \qquad H = Rp_* G = \mathsf{S}\mathbf{R}(F). \tag{3.6.2}$$

Consider the two adjunction morphisms

$$\eta_F : F \longrightarrow H, \qquad \bar{\epsilon}_F : H \longrightarrow F, \tag{3.6.3}$$

where η_F is the unit of $\mathbf{R} \dashv \mathsf{S}$ and $\bar{\epsilon}_F$ is the counit of $\mathsf{S} \dashv \mathbf{R}$. Equivalently, $\bar{\epsilon}_F$ is the Verdier-dual mate of the Radon unit. Thus $\bar{\epsilon}_F \eta_F$ is the cross-composite of the two adjunctions and is not a composite occurring in either triangle identity.

Lemma 3.7 (The Radon sign)**.** *With the absolute-purity identifications $h^! \simeq h^*[2d](d)$ and $p^! \simeq p^*[2d](d)$, we have*

$$\bar{\epsilon}_F \eta_F = (-1)^d \operatorname{id}_F. \tag{3.7.1}$$

Thus $r_F = (-1)^d \bar{\epsilon}_F$ satisfies $r_F \eta_F = \operatorname{id}_F$.

Proof. Put $A = p^* F$. We denote by $u_p : \mathrm{id} \to Rp_* p^*$ the unit of $p^* \dashv Rp_*$, by $u_h^! : \mathrm{id} \to h^! Rh_*$ the unit of $Rh_! = Rh_* \dashv h^!$, by $\epsilon_h : h^* Rh_* \to \mathrm{id}$ the counit of $h^* \dashv Rh_*$, and by $\mathrm{Tr}_p : Rp_* p^! \to \mathrm{id}$ the trace map. Unfolding the two Radon adjunctions, together with their shifts and twists, we identify $\bar{\epsilon}_F \eta_F$ with the following composite:

$$\begin{aligned} &F \xrightarrow{u_p} Rp_* A \xrightarrow{Rp_* u_h^!} Rp_* h^! Rh_* A \\ &\xrightarrow{Rp_* \operatorname{pur}_h^{-1}} Rp_* h^* Rh_* A[2d](d) \xrightarrow{Rp_* \epsilon_h [2d](d)} Rp_* A[2d](d) \xrightarrow{\mathrm{Tr}_p} F. \end{aligned} \tag{3.7.2}$$

Here $\operatorname{pur}_h : h^*(-)[2d](d) \xrightarrow{\sim} h^!(-)$ denotes the purity isomorphism; in the last arrow we use the identification $A[2d](d) = p^! F$.

We compute the middle part of (3.7.2) by the Euler transformation. Let $f : Y \to Z$ be a smooth proper morphism of relative dimension d and let M be a finite-Tor-dimension complex on Y. Then the composite

$$M \xrightarrow{u_f^!} f^! Rf_* M \xrightarrow{\operatorname{pur}_f^{-1}} f^* Rf_* M[2d](d) \xrightarrow{\epsilon_f [2d](d)} M[2d](d) \tag{3.7.3}$$

is cup product by the Euler class $e(T_f) = c_d(T_f)$. To see this, express the unit and the counit by means of the relative diagonal

$$\delta_f : Y \hookrightarrow Y \times_Z Y.$$

The normal bundle of δ_f is canonically T_f. The absolute-purity self-intersection formula then identifies (3.7.3) with cup product by $c_d(T_f)$ [SGA5, Exposé VII, §4].

Applying this assertion to $f = h$ and $M = A$, and then applying the projection formula to (3.7.2), we obtain

$$\bar{\epsilon}_F \eta_F = \big(p_* c_d(T_h)\big)\, \mathrm{id}_F, \tag{3.7.4}$$

where p_* denotes the Gysin trace on cohomology.

It remains to evaluate $p_* c_d(T_h)$. Let $y = [L] \in P$ be a geometric point and put

$$B = p^{-1}(y) = \mathbf{P}((V/L)^\vee) \simeq \mathbf{P}^d.$$

Let $\mathcal{U}$ be the universal hyperplane subbundle on B, defined by

$$0 \longrightarrow \mathcal{U} \longrightarrow (V/L) \otimes \mathcal{O}_B \longrightarrow \mathcal{O}_B(1) \longrightarrow 0. \tag{3.7.5}$$

At the point of B corresponding to a hyperplane $H \subset V$ containing L, the tangent space of the fiber of h is $\mathrm{Hom}(L, H/L)$. Hence

$$T_h|_B \simeq L^\vee \otimes \mathcal{U}. \tag{3.7.6}$$

The factor $L^\vee$ is a constant line on B and has trivial Chern class. If $a = c_1(\mathcal{O}_B(1))$, then (3.7.5) gives

$$c_d(T_h|_B) = c_d(\mathcal{U}) = (-1)^d a^d, \qquad \mathrm{Tr}_B(a^d) = 1. \tag{3.7.7}$$

By proper base change for the trace, we have $p_* c_d(T_h) = (-1)^d$ in $H^0(P, \Lambda)$. Substituting this equality into (3.7.4), we obtain (3.7.1). $\square$

Since $r_F \eta_F = \mathrm{id}_F$, conjugation gives morphisms of endomorphism kernels

$$U_F : \mathscr{H}_F \longrightarrow \mathscr{H}_H, \qquad a \longmapsto \eta_F \circ a \circ r_F, \tag{3.7.8}$$

$$V_F : \mathscr{H}_H \longrightarrow \mathscr{H}_F, \qquad b \longmapsto r_F \circ b \circ \eta_F. \tag{3.7.9}$$

More explicitly, U_F is precomposition with $\mathrm{pr}_1^* r_F$ followed by postcomposition with $\mathrm{pr}_2^! \eta_F$; V_F uses $\mathrm{pr}_1^* \eta_F$ and $\mathrm{pr}_2^! r_F$ in the reverse order.

Theorem 3.8 (Specialized Radon retraction)**.** *Let*

$$\beta_G = \beta_{G,G} : R\varrho_* \nu_C(\mathscr{H}_G) \xrightarrow{\sim} \nu_{\Delta_P}(\mathscr{H}_H) \tag{3.8.1}$$

be the proper-specialization isomorphism of proposition 3.5. Define

$$U_F^{\mathrm{sp}} = \beta_G^{-1} \circ \nu_{\Delta_P}(U_F), \tag{3.8.2}$$

$$V_F^{\mathrm{sp}} = \nu_{\Delta_P}(V_F) \circ \beta_G. \tag{3.8.3}$$

Then

$$\nu_{\Delta_P}(\mathscr{H}_F) \xrightarrow{U_F^{\mathrm{sp}}} R\varrho_* \nu_C(\mathscr{H}_G) \xrightarrow{V_F^{\mathrm{sp}}} \nu_{\Delta_P}(\mathscr{H}_F), \tag{3.8.4}$$

and the composite is the identity. The two arrows are the specialized maps on endomorphism kernels induced by the Radon unit η_F and the normalized opposite counit r_F, transported through proper specialization.

Proof. Since $r_F\eta_F = \mathrm{id}_F$, on $P\times P$ we have

$$V_FU_F(a) = r_F\eta_F\, a\, r_F\eta_F = a.$$

Using (3.8.2)–(3.8.3), functoriality of Verdier specialization gives

$$\begin{aligned} V_F^{\mathrm{sp}}U_F^{\mathrm{sp}} &= \nu_{\Delta_P}(V_F)\,\beta_G\beta_G^{-1}\,\nu_{\Delta_P}(U_F) \\ &= \nu_{\Delta_P}(V_FU_F) = \nu_{\Delta_P}(\mathrm{id}_{\mathscr{H}_F}) = \mathrm{id}_{\nu_{\Delta_P}(\mathscr{H}_F)}\,. \end{aligned}$$

This proves that (3.8.4) is a retraction.

It remains to identify the two arrows as specialized cohomological correspondences. Put

$$Z_h = Q\times_{P^\vee} Q, \qquad a_i = p\circ \mathrm{pr}_i : Z_h \longrightarrow P \quad (i=1,2), \tag{3.8.5}$$

and let

$$\delta_h : Q \hookrightarrow Z_h, \qquad q\longmapsto (q,q), \tag{3.8.6}$$

be the relative diagonal of h. By proper base change for the cartesian square defining Z_h, we have a canonical isomorphism

$$H = \mathsf{S}\mathbf{R}(F) \xrightarrow{\sim} Ra_{2*}a_1^*F[2d](d). \tag{3.8.7}$$

Under (3.8.7), the unit η_F is the coevaluation cell supported on δ_h. More explicitly, it is the following composition, already used in (3.7.2):

$$F \xrightarrow{u_p} Rp_*p^*F \xrightarrow{Rp_*u_h^!} Rp_*h^!Rh_*p^*F = H, \tag{3.8.8}$$

where the equality is induced by purity. In particular, the unit occurring here is $u_h^! : \mathrm{id}\to h^!Rh_*$ for the adjunction $Rh_! = Rh_* \dashv h^!$, rather than the unit of $h^*\dashv Rh_*$. Its Verdier-dual evaluation cell, multiplied by $(-1)^d$, is r_F. Hence the maps U_F and V_F on the internal endomorphism objects are induced by the pairs (r_F,η_F) and (η_F, r_F), respectively.

By its construction in proposition 3.5, β_G is obtained by applying Lu–Zheng specialization to these cells and then using the proper morphism Π. The binaturality of β_G and its compatibility with composition and duality give the commutative diagrams

$$\begin{tikzcd} \nu_{\Delta_P}\mathscr{H}_F \arrow[r, "\nu_{\Delta_P}(U_F)"] \arrow[d, equal] & \nu_{\Delta_P}\mathscr{H}_H \arrow[d, "\beta_G^{-1}"] \\ \nu_{\Delta_P}\mathscr{H}_F \arrow[r, "U_F^{\mathrm{sp}}"'] & R\varrho_*\nu_C\mathscr{H}_G \end{tikzcd} \qquad \begin{tikzcd} \nu_{\Delta_P}\mathscr{H}_H \arrow[r, "\nu_{\Delta_P}(V_F)"] \arrow[d, "\beta_G^{-1}"'] & \nu_{\Delta_P}\mathscr{H}_F \arrow[d, equal] \\ R\varrho_*\nu_C\mathscr{H}_G \arrow[r, "V_F^{\mathrm{sp}}"'] & \nu_{\Delta_P}\mathscr{H}_F. \end{tikzcd} \tag{3.8.9}$$

Indeed, [LZ22, Construction 3.3] preserves the two compositions defining U_F and V_F, and [LZ22, Proposition 3.12] identifies specialization before and after proper pushforward. Applying the resulting proper-pushforward diagram to the cell (3.8.8) gives the first square. Applying the same argument to the Verdier-dual cell for $\mathbb{D}F$, using the factor-exchange identification of endomorphism kernels, and multiplying by the sign $(-1)^d$ of lemma 3.7, we obtain the second square. This proves the asserted cohomological-correspondence interpretation of U_F^{sp} and V_F^{sp} [LZ22, Construction 3.3 and Proposition 3.12]. □

4. Fourier localization on the collision normal bundle

4.1. We first prove a Fourier vanishing lemma for complexes which are constant in a vector-bundle direction. Consider the notation in 3.1 and 3.2. Put

$$N = C\times_P TP, \qquad \varrho : N\to TP, \qquad \mathscr{K}_G = \nu_C(\mathscr{H}_G). \tag{4.1.1}$$

For open subsets $W_1, W_2\subset Q$, put

$$C(W_1,W_2) = C\cap (W_1\times W_2), \qquad N(W_1,W_2) = C(W_1,W_2)\times_P TP.$$

Lemma 4.2 (Constant vector direction)**.** *Let $e : V \to S$ be a vector bundle of positive rank r and let $a : T \to S$ a separated morphism of finite type. Let $M \in D^b_{ctf}(T, \Lambda)$. Consider the following cartesian square*

$$\begin{tikzcd} T \times_S V \arrow[r, "\mathrm{pr}_T"] \arrow[d, "a_V"'] & T \arrow[d, "a"] \\ V \arrow[r, "e"'] & S \end{tikzcd} \tag{4.2.1}$$

Then we have

$$\mathfrak{F}_{\psi,V/S} Ra_{V!} \,\mathrm{pr}_T^* M\big|_{V^{\vee\times}} = 0. \tag{4.2.2}$$

If a is proper, the same assertion holds with $R(-)_*$ since $Ra_{V!} = Ra_{V*}$.

Proof. By proper base change theorem, there is a canonical isomorphism

$$Ra_{V!} \,\mathrm{pr}_T^* M \xrightarrow{\sim} e^* Ra_! M. \tag{4.2.3}$$

Let $0 \neq \xi \in V^\vee_{\bar{s}}$ be a nonzero geometric covector over a geometric point $\bar{s} \to S$. Then we have

$$\begin{aligned} \big(\mathfrak{F}_{\psi,V/S} Ra_{V!} \,\mathrm{pr}_T^* M\big)_\xi &\simeq R\Gamma_c\big(V_{\bar{s}}, (Ra_! M)_{\bar{s}} \otimes^L_\Lambda \mathcal{L}_\psi(\langle v, \xi\rangle)\big) \\ &\simeq (Ra_! M)_{\bar{s}} \otimes^L_\Lambda R\Gamma_c(V_{\bar{s}}, \mathcal{L}_\psi(\langle v, \xi\rangle)) \,. \end{aligned} \tag{4.2.4}$$

Since $\xi \neq 0$, we can choose linear coordinates on $V_{\bar{s}} \simeq \mathbf{A}^r$ such that $\langle v, \xi\rangle = cv_1$ for some $c \in \bar{k}^\times$. The Künneth formula gives

$$R\Gamma_c(V_{\bar{s}}, \mathcal{L}_\psi(\langle v, \xi\rangle)) \simeq R\Gamma_c(\mathbf{A}^1, \mathcal{L}_\psi(cv_1)) \otimes^L_\Lambda R\Gamma_c(\mathbf{A}^{r-1}, \Lambda). \tag{4.2.5}$$

For $c \neq 0$, we have $R\Gamma_c(\mathbf{A}^1, \mathcal{L}_\psi(cv_1)) = 0$. Thus (4.2.5) and (4.2.4) vanish for every nonzero ξ. This proves (4.2.2). □

4.3. We recall the equivariance condition for constructible sheaves. Let $a : T \to S$ be a proper morphism of schmes. Let $V \to S$ and $E \to T$ be vector bundles. Put $V_T = T \times_S V$. Let $\alpha : E \to V_T$ be a linear homomorphism and consider

$$\mu : E \times_T V_T \longrightarrow V_T, \qquad (u, v) \longmapsto v + \alpha(u), \qquad q : E \times_T V_T \longrightarrow V_T, \quad (u, v) \longmapsto v.$$

An E-equivariant structure on $K \in D^b_c(V_T, \Lambda)$ is an isomorphism

$$\theta : \mu^* K \xrightarrow{\sim} q^* K \tag{4.3.1}$$

with the following unit and cocycle properties. If

$$e : V_T \longrightarrow E \times_T V_T, \qquad v \longmapsto (0, v),$$

then

$$e^* \theta = \mathrm{id}_K \,. \tag{4.3.2}$$

On $E \times_T E \times_T V_T$, let

$$M(u_1, u_2, v) = (u_1 + u_2, v), \qquad A(u_1, u_2, v) = (u_1, v + \alpha(u_2)),$$

and let $p_{23}(u_1, u_2, v) = (u_2, v)$. Thus M, A, p_{23} take values in $E \times_T V_T$. The cocycle condition is

$$M^* \theta = p_{23}^* \theta \circ A^* \theta. \tag{4.3.3}$$

Lemma 4.4 (Equivariant Fourier support)**.** *Let $a : T \to S$ be a proper morphism of schmes. Let $V \to S$ and $E \to T$ be vector bundles. Let $\alpha : E \to V_T$ be a linear homomorphism and $\alpha^\vee : V_T^\vee \to E^\vee$ the dual homomorphism. Suppose that $K \in D^b_c(V_T, \Lambda)$ is E-equivariant. Write*

$$b = \mathrm{pr}_2 : V_T = T \times_S V \longrightarrow V, \qquad b^\vee = \mathrm{pr}_2 : V_T^\vee = T \times_S V^\vee \longrightarrow V^\vee,$$

Then we have

$$\operatorname{supp}\mathfrak{F}_{\psi,V/S}Rb_*K \subset b^\vee\big(Z(\alpha^\vee)\big), \tag{4.4.1}$$

where

$$Z(\alpha^\vee) = \ker(\alpha^\vee) = \{(t,\xi) \in V_T^\vee \mid \alpha_t^\vee(\xi) = 0\} \tag{4.4.2}$$

is regarded as a closed subscheme of $V_T^\vee$.

Proof. By [Sai26, Proposition 1.1.8], there is a canonical isomorphism

$$\mathfrak{F}_{\psi,V/S}Rb_*K \xrightarrow{\sim} Rb_*^\vee \mathfrak{F}_{\psi,V_T/T}K. \tag{4.4.3}$$

Now we prove

$$\operatorname{supp}\mathfrak{F}_{\psi,V_T/T}K \subset Z(\alpha^\vee). \tag{4.4.4}$$

It is enough to check the vanishing at geometric points. Let $(\bar{t},\xi)$ be a geometric point of $V_T^\vee$ lying above $\bar{s} \to S$. Assume that

$$\alpha_{\bar{t}}^\vee(\xi) \neq 0. \tag{4.4.5}$$

Choose a one-dimensional subspace $L \subset E_{\bar{t}}$ on which this linear form is nonzero. Then $\alpha_{\bar{t}}|_L$ is injective, and

$$W = \alpha_{\bar{t}}(L) \subset V_{\bar{s}}$$

is a line satisfying $\xi|_W \neq 0$.

Restricting (4.3.1) and its unit and cocycle conditions to L gives an W-equivariant structure on $K_{\bar{t}}$. We define this structure explicitly. Choose a linear splitting

$$V_{\bar{s}} = W \oplus V'.$$

Let $i : V' \hookrightarrow V_{\bar{s}}$ be the second summand and put $K' = i^*K_{\bar{t}}$. The action map

$$h : W \times V' \xrightarrow{\sim} V_{\bar{s}}, \qquad (w,v') \longmapsto w + i(v'),$$

is an isomorphism. Pulling the isomorphism (4.3.1) back by the map

$$W \times V' \longrightarrow W \times V_{\bar{s}}, \qquad (w,v') \longmapsto (w, i(v')),$$

gives

$$h^*K_{\bar{t}} \xrightarrow{\sim} \operatorname{pr}_{V'}^* K'.$$

Transporting this isomorphism through h yields

$$K_{\bar{t}} \simeq \operatorname{pr}_{V'}^* K' \qquad \text{under } V_{\bar{s}} = W \oplus V'. \tag{4.4.6}$$

Thus $K_{\bar{t}}$ is pulled back from the quotient $V_{\bar{s}}/W \simeq V'$. The unit and cocycle conditions ensure that this identification is the normalized descent datum associated with the W-torsor $V_{\bar{s}} \to V_{\bar{s}}/W$ and is independent of the presentation of a translation as a sum of elements of W.

The stalk formula for the relative Fourier transform gives

$$\begin{aligned}\big(\mathfrak{F}_{\psi,V_T/T}K\big)_{(\bar{t},\xi)} &\simeq R\Gamma_c\big(V_{\bar{s}}, K_{\bar{t}} \otimes_\Lambda^L \mathcal{L}_\psi(\langle v,\xi\rangle)\big) \\ &\simeq R\Gamma_c(W, \mathcal{L}_\psi(\xi|_W)) \otimes_\Lambda^L R\Gamma_c\big(V', K' \otimes_\Lambda^L \mathcal{L}_\psi(\xi|_{V'})\big)\,,\end{aligned} \tag{4.4.7}$$

where the second isomorphism uses (4.4.6) and the Künneth formula. Because $\xi|_W$ is a nonzero linear form, we have

$$R\Gamma_c(W, \mathcal{L}_\psi(\xi|_W)) = 0.$$

Consequently the stalk in (4.4.7) vanishes. Since this holds at every geometric point outside $Z(\alpha^\vee)$, it proves (4.4.4).

Finally, (4.4.3) and (4.4.4) imply

$$\operatorname{supp}\mathfrak{F}_{\psi,V/S}Rb_*K \subset b^\vee\big(\operatorname{supp}\mathfrak{F}_{\psi,V_T/T}K\big) \subset b^\vee\big(Z(\alpha^\vee)\big).$$

The last image is closed because $b^\vee$ is proper. This proves (4.4.1). □

4.5. We now apply the preceding equivariant Fourier-support lemma to the incidence correspondence. Since $Q \subseteq P \times P^\vee$, we have $T_{Q/P^\vee} \subseteq Q \times_P TP$. On $C = Q \times_P Q$, we define

$$\alpha : \mathcal{E} := \mathrm{pr}_1^* T_{Q/P^\vee} \oplus \mathrm{pr}_2^* T_{Q/P^\vee} \to C \times_P TP, \qquad \alpha(u_1, u_2) = dp(u_1) - dp(u_2) \in TP. \tag{4.5.1}$$

It acts on $N = C \times_P TP$ by translation in the TP-coordinate. Recall the notation $p : Q \to P$ and $h : Q \to P^\vee$. Let $F \in D^b_{ctf}(P, \Lambda)$. Put $A = \mathbf{R}(F)$ and $G = h^* A[n-1](n-1)$.

Lemma 4.6 (Incidence equivariance)**.** *The complex $\mathscr{K}_G = \nu_C(\mathscr{H}_G)$ is $\mathcal{E}$-equivariant. At $(q_1, q_2) = (y, H_1, H_2)$ one has*

$$\operatorname{im} \alpha_{q_1,q_2} = T_y H_1 + T_y H_2. \tag{4.6.1}$$

Consequently α is surjective off Δ_Q. For $0 \neq \xi \in T_y^* P$, we have

$$\alpha^\vee_{q_1,q_2}(\xi) = 0 \Longleftrightarrow q_1 = q_2 = (y, H_\xi), \qquad T_y H_\xi = \ker \xi. \tag{4.6.2}$$

Proof. We first construct the equivariance intrinsically. Put

$$Y = Q \times Q, \qquad B = P^\vee \times P^\vee, \qquad g = h \times h : Y \longrightarrow B, \qquad A = \mathbf{R}(F).$$

The morphism h is smooth of relative dimension d. Since $G = h^* A[d](d)$, by smooth purity and (3.5.4) we have canonical isomorphisms

$$\begin{aligned} \mathbb{D}_Q G &\simeq h^* \mathbb{D}_{P^\vee} A[d], \\ \mathscr{H}_G &\simeq \mathbb{D}_Q G \boxtimes G \simeq g^* (\mathbb{D}_{P^\vee} A \boxtimes A)\,[2d](d). \end{aligned} \tag{4.6.3}$$

Thus $\mathscr{H}_G$, with the indicated shifts and twists, is canonically pulled back from B.

Consider the relative pair groupoid

$$\Gamma = Y \times_B Y \rightrightarrows Y, \qquad s = \mathrm{pr}_2, \quad t = \mathrm{pr}_1, \quad e = \Delta_{Y/B}. \tag{4.6.4}$$

Since $g \circ s = g \circ t$, (4.6.3) gives a canonical isomorphism

$$\epsilon : t^* \mathscr{H}_G \xrightarrow{\sim} s^* \mathscr{H}_G. \tag{4.6.5}$$

Under (4.6.3), this is the identity between the two pullbacks from B. Hence it is normalized along e and satisfies the groupoid cocycle.

We deform the groupoid action. Put

$$\mathscr{G} = \mathscr{D}_{e(Y)}(\Gamma), \qquad \mathscr{Y} = \mathscr{D}_C(Y),$$

and let $\rho : \mathscr{Y} \to Y \times \mathbf{A}^1$ be the structural morphism. The source, target, unit, inverse, and multiplication maps induce morphisms of deformation spaces. We shall verify below that they make $\mathscr{G}$ a smooth groupoid over $Y \times \mathbf{A}^1$. Its generic and special fibers are

$$\mathscr{G}_{\mathbf{G}_m} = \Gamma \times \mathbf{G}_m, \qquad \mathscr{G}_0 = N_{e(Y)/\Gamma} \simeq T_{Y/B}, \tag{4.6.6}$$

and the group law on the special fiber is addition in $T_{Y/B}$.

We first construct the comparison with the action space. The inclusion $e(C) \subset e(Y)$ and the identity $s \circ e = \mathrm{id}_Y$ give morphisms of pairs

$$(\Gamma, e(C)) \longrightarrow (\Gamma, e(Y)), \qquad s : (\Gamma, e(C)) \longrightarrow (Y, C).$$

The induced morphisms of deformation spaces have the same composite to $Y \times \mathbf{A}^1$. Hence the universal property of the fiber product gives a canonical morphism

$$\beta : \mathscr{D}_{e(C)}(\Gamma) \longrightarrow \mathscr{G} \underset{s_{\mathscr{G}}, Y \times \mathbf{A}^1, \rho}{\times} \mathscr{Y}. \tag{4.6.7}$$

We show that β is an isomorphism. Its inverse will identify the action space with $\mathscr{D}_{e(C)}(\Gamma)$.

Let $I = I_{e(Y)} \subset \mathscr{O}_\Gamma$ and $J = I_C \subset \mathscr{O}_Y$. Scheme-theoretically,

$$I_{e(C)} = I + s^*J \cdot \mathscr{O}_\Gamma. \tag{4.6.8}$$

Indeed, $e(C)$ is the fiber product of $e(Y) \subset \Gamma$ and $C \subset Y$ with respect to s, and $s|_{e(Y)}$ is inverse to e. Write

$$\mathcal{R}_I^{\mathrm{ext}} = \mathscr{O}_\Gamma[\tau, I\tau^{-1}], \qquad \mathcal{R}_J^{\mathrm{ext}} = \mathscr{O}_Y[\tau, J\tau^{-1}]$$

for the extended Rees algebras. On coordinate algebras, β is induced by the canonical homomorphism

$$\mathcal{R}_I^{\mathrm{ext}} \underset{s^{-1}\mathscr{O}_Y[\tau]}{\otimes} s^{-1}\mathcal{R}_J^{\mathrm{ext}} \longrightarrow \mathcal{R}_{I+s^*J}^{\mathrm{ext}}, \qquad \frac{a}{\tau^m} \otimes \frac{b}{\tau^n} \longmapsto \frac{a\, s^*b}{\tau^{m+n}}. \tag{4.6.9}$$

This homomorphism, and hence β, is defined without choosing coordinates.

We verify that (4.6.9) is an isomorphism. Over $\tau \neq 0$, both sides of (4.6.7) are canonically $\Gamma \times \mathbf{G}_m$, and β is the identity. Every point of the special fiber lies over $e(Y)$. Since s is smooth and e is a section, étale locally on Γ in a neighborhood of any point of $e(Y)$ there is an étale morphism, over Y, to $Y \times \mathbf{A}^r$ carrying e to the zero section, where $r = \dim(Y/B)$. Formation of deformation spaces and the morphism β commutes with this étale base change. Hence it suffices to compute in the standard model

$$\Gamma = \operatorname{Spec} A[x_1, \ldots, x_r], \qquad Y = \operatorname{Spec} A, \qquad I = (x_1, \ldots, x_r),$$

with $J \subset A$ the ideal defining C.

Put $u_i = x_i\tau^{-1}$. Since $x_i = \tau u_i$, the coordinate algebra of the fiber product in (4.6.7) is

$$\begin{aligned} &A[x_1, \ldots, x_r, \tau, x_1\tau^{-1}, \ldots, x_r\tau^{-1}] \underset{A[\tau]}{\otimes} A[\tau, J\tau^{-1}] \\ &\simeq A[\tau, J\tau^{-1}, u_1, \ldots, u_r]. \end{aligned} \tag{4.6.10}$$

On the other hand, (4.6.8) gives

$$A[x_1, \ldots, x_r, \tau, (J, x_1, \ldots, x_r)\tau^{-1}] \simeq A[\tau, J\tau^{-1}, u_1, \ldots, u_r], \tag{4.6.11}$$

which is the coordinate algebra of $\mathscr{D}_{e(C)}(\Gamma)$. Under (4.6.10) and (4.6.11), the homomorphism (4.6.9) is the identity. It follows that β is an isomorphism in an étale neighborhood of every point of the special fiber. Together with the calculation over $\tau \neq 0$, this proves that β is globally an isomorphism. Notice that the coordinates $x_1, \ldots, x_r$ are used only to verify the canonical morphism β; no coordinate-dependent isomorphisms are glued.

We next verify compatibility with composable arrows. Put

$$M = \Gamma \underset{s,Y,t}{\times} \Gamma, \qquad Z = e(Y) \underset{Y}{\times} e(Y) \subset M.$$

Thus M is the scheme of composable pairs of arrows and $Z \simeq Y$ is the subscheme of pairs of identity arrows. For the relative pair groupoid, we have

$$M \simeq Y \times_B Y \times_B Y,$$

and Z is the relative triple diagonal. Let $p_1, p_2 : M \to \Gamma$ be the projections. Since each p_i maps Z into $e(Y)$, they induce morphisms

$$\mathscr{D}_Z(M) \longrightarrow \mathscr{G}.$$

Since $s \circ p_1 = t \circ p_2$, the two morphisms have the same image in $Y \times \mathbf{A}^1$. Hence they define a canonical morphism

$$\gamma : \mathscr{D}_Z(M) \longrightarrow \mathscr{G} \underset{s_{\mathscr{G}}, Y \times \mathbf{A}^1, t_{\mathscr{G}}}{\times} \mathscr{G}. \tag{4.6.12}$$

We show that γ is an isomorphism. Over $\tau \neq 0$, both sides of (4.6.12) are canonically $M \times \mathbf{G}_m$. Along the special fiber, we work étale locally over B and choose relative coordinates. In the standard model, write a point of M as $(\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2)$, with $\mathbf{x}_i = (x_{i,1}, \ldots, x_{i,r})$. The two arrows are $(\mathbf{x}_0, \mathbf{x}_1)$ and $(\mathbf{x}_1, \mathbf{x}_2)$, and the ideal of Z is generated by the two r-tuples

$$\mathbf{x}_0 - \mathbf{x}_1, \qquad \mathbf{x}_1 - \mathbf{x}_2.$$

Set

$$\mathbf{u} = (\mathbf{x}_0 - \mathbf{x}_1)\tau^{-1}, \qquad \mathbf{v} = (\mathbf{x}_1 - \mathbf{x}_2)\tau^{-1}.$$

If A is the coordinate algebra of the corresponding affine chart of B, then the coordinate algebra of $\mathscr{D}_Z(M)$ is

$$\frac{A[\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2, \tau, \mathbf{u}, \mathbf{v}]}{(\mathbf{x}_0 - \mathbf{x}_1 - \tau\mathbf{u}, \mathbf{x}_1 - \mathbf{x}_2 - \tau\mathbf{v})} \simeq A[\mathbf{x}_2, \tau, \mathbf{u}, \mathbf{v}]. \tag{4.6.13}$$

Here an equation between bold symbols denotes the r component equations.

The first factor $\mathscr{G}$, corresponding to $(\mathbf{x}_0, \mathbf{x}_1)$, has coordinate algebra

$$\frac{A[\mathbf{x}_0, \mathbf{x}_1, \tau, \mathbf{u}]}{(\mathbf{x}_0 - \mathbf{x}_1 - \tau\mathbf{u})},$$

and the second factor, corresponding to $(\mathbf{x}_1, \mathbf{x}_2)$, has coordinate algebra

$$\frac{A[\mathbf{x}_1, \mathbf{x}_2, \tau, \mathbf{v}]}{(\mathbf{x}_1 - \mathbf{x}_2 - \tau\mathbf{v})}.$$

Their fiber product in (4.6.12) identifies the middle coordinate $\mathbf{x}_1$. Its coordinate algebra is therefore

$$\begin{aligned} &\frac{A[\mathbf{x}_0, \mathbf{x}_1, \tau, \mathbf{u}]}{(\mathbf{x}_0 - \mathbf{x}_1 - \tau\mathbf{u})} \underset{A[\mathbf{x}_1,\tau]}{\otimes} \frac{A[\mathbf{x}_1, \mathbf{x}_2, \tau, \mathbf{v}]}{(\mathbf{x}_1 - \mathbf{x}_2 - \tau\mathbf{v})} \\ &\simeq \frac{A[\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2, \tau, \mathbf{u}, \mathbf{v}]}{(\mathbf{x}_0 - \mathbf{x}_1 - \tau\mathbf{u}, \mathbf{x}_1 - \mathbf{x}_2 - \tau\mathbf{v})}. \end{aligned} \tag{4.6.14}$$

Under (4.6.13) and (4.6.14), the homomorphism defining γ is the identity. Thus γ is an isomorphism $\acute{e}$tale locally along the special fiber, and therefore globally.

Let $m : M \to \Gamma$ be the ordinary groupoid multiplication. In the above coordinates it is

$$m(\mathbf{x}_0, \mathbf{x}_1, \mathbf{x}_2) = (\mathbf{x}_0, \mathbf{x}_2),$$

and

$$\frac{\mathbf{x}_0 - \mathbf{x}_2}{\tau} = \frac{\mathbf{x}_0 - \mathbf{x}_1}{\tau} + \frac{\mathbf{x}_1 - \mathbf{x}_2}{\tau} = \mathbf{u} + \mathbf{v}. \tag{4.6.15}$$

It follows that m induces a morphism $\mathscr{D}_Z(M) \to \mathscr{G}$; after identifying its source by γ, it is the multiplication on $\mathscr{G}$, and its special fiber is addition on $T_{Y/B}$. The unit specializes to the zero section and inversion to $\mathbf{u} \mapsto -\mathbf{u}$. Associativity, the unit identities, and the inverse identity follow from the corresponding identities before deformation, or, equivalently, from the induced homomorphisms of Rees algebras. By [Sai26, Lemma 1.2.3(2)], the source map is smooth; the target map is smooth by inversion.

The target map $t : \Gamma \to Y$ carries $e(C)$ to C. It therefore induces a morphism of deformations

$$\widetilde{\mu} : \mathscr{D}_{e(C)}(\Gamma) \longrightarrow \mathscr{Y}. \tag{4.6.16}$$

We use (4.6.16) to define the action. The identity morphism of Γ, regarded as a morphism of pairs $(\Gamma, e(C)) \to (\Gamma, e(Y))$, induces

$$\widetilde{\pi} : \mathscr{D}_{e(C)}(\Gamma) \longrightarrow \mathscr{G}.$$

The source map $s : (\Gamma, e(C)) \to (Y, C)$ similarly induces

$$\widetilde{q} = \mathscr{D}(s) : \mathscr{D}_{e(C)}(\Gamma) \longrightarrow \mathscr{Y}.$$

By construction, the canonical isomorphism (4.6.7) is

$$\beta = (\widetilde{\pi}, \widetilde{q}) : \mathscr{D}_{e(C)}(\Gamma) \xrightarrow{\sim} \mathscr{G} \underset{s_{\mathscr{G}}, Y\times\mathbf{A}^1, \rho}{\times} \mathscr{Y}.$$

Hence the action morphism on the standard action space is

$$a = \widetilde{\mu} \circ \beta^{-1} : \mathscr{G} \underset{s_{\mathscr{G}}, Y\times\mathbf{A}^1, \rho}{\times} \mathscr{Y} \longrightarrow \mathscr{Y}. \tag{4.6.17}$$

Through β, we also call $\widetilde{\mu}$ the action morphism.

We verify the action identities. By functoriality for the target map, the following diagram is commutative:

$$\begin{tikzcd} \mathscr{D}_{e(C)}(\Gamma) \arrow[r, "\widetilde{\mu}"] \arrow[d, "\widetilde{\pi}"'] & \mathscr{Y} \arrow[d, "\rho"] \\ \mathscr{G} \arrow[r, "t_{\mathscr{G}}"'] & Y \times \mathbf{A}^1. \end{tikzcd}$$

Therefore

$$\rho \circ a = t_{\mathscr{G}} \circ \mathrm{pr}_1, \tag{4.6.18}$$

so $a(\eta, x)$ lies over the target of η.

For the unit identity, e is a morphism of pairs $(Y, C) \to (\Gamma, e(C))$ and therefore induces

$$\widetilde{e} : \mathscr{Y} \longrightarrow \mathscr{D}_{e(C)}(\Gamma).$$

The identities $s \circ e = t \circ e = \mathrm{id}_Y$ give

$$\beta \circ \widetilde{e} = (e_{\mathscr{G}} \circ \rho, \mathrm{id}_{\mathscr{Y}}), \qquad \widetilde{\mu} \circ \widetilde{e} = \mathrm{id}_{\mathscr{Y}}.$$

By (4.6.17), this gives

$$a(e_{\mathscr{G}}(\rho(x)), x) = x. \tag{4.6.19}$$

For associativity, consider the scheme of triples (η_1, η_2, x) with

$$s_{\mathscr{G}}(\eta_2) = \rho(x), \qquad s_{\mathscr{G}}(\eta_1) = t_{\mathscr{G}}(\eta_2).$$

The composable-arrow comparison (4.6.12) and the action-space comparison (4.6.7) identify the two action diagrams with the deformations of the corresponding diagrams for the relative pair groupoid. Thus functoriality gives

$$a(\eta_1, a(\eta_2, x)) = a(m_{\mathscr{G}}(\eta_1, \eta_2), x). \tag{4.6.20}$$

The identities (4.6.18), (4.6.19), and (4.6.20) prove that a is a left action of $\mathscr{G} \rightrightarrows Y \times \mathbf{A}^1$ on $\mathscr{Y}$ with anchor ρ.

We shall also use that both action-space maps are smooth. Under β, $\widetilde{q}$ is the second projection and is therefore the base change of the smooth source map $s_{\mathscr{G}}$. The involution

$$(\eta, x) \longmapsto (\eta^{-1}, a(\eta, x))$$

of the action space exchanges the second projection with a. Hence a, or equivalently $\widetilde{\mu}$, is smooth.

We now compute the special action. At $c \in C$, the differential of s and the unit e give a canonical splitting

$$T_{\Gamma, e(c)} \simeq T_{Y,c} \oplus T_{Y/B,c}.$$

For (v, u) in this splitting one has

$$ds(v, u) = v, \qquad dt(v, u) = v + u.$$

After quotienting by $T_c C$, the special fibers of $\widetilde{q}$ and $\widetilde{\mu}$ are therefore

$$q : T_{Y/B}|_C \times_C N_{C/Y} \longrightarrow N_{C/Y}, \quad (u, \bar{v}) \longmapsto \bar{v}, \qquad \mu(u, \bar{v}) = \bar{v} + \bar{u}, \tag{4.6.21}$$

where $\bar{u}$ is the image of u under $T_{Y/B}|_C \to N_{C/Y}$.

It remains to specialize the generic equivariance. We first describe the two smooth-base-change isomorphisms. Let

$$\mathscr{T} = \operatorname{Spec} \mathcal{O}^h_{\mathbf{A}^1,0}$$

be the henselian trait at the origin, with closed point s and chosen geometric generic point η_0. Put

$$\mathscr{Y}_{\mathscr{T}} = \mathscr{Y} \times_{\mathbf{A}^1} \mathscr{T}, \qquad \mathscr{A}_{\mathscr{T}} = \mathscr{D}_{e(C)}(\Gamma) \times_{\mathbf{A}^1} \mathscr{T}.$$

Write $j_{\mathscr{Y}}, i_{\mathscr{Y}}$ and $j_{\mathscr{A}}, i_{\mathscr{A}}$ for the generic- and special-fiber morphisms, respectively. The generic and special fibers are canonically identified as

$$\mathscr{Y}_{\eta_0} = Y \times \eta_0, \qquad \mathscr{A}_{\eta_0} = \Gamma \times \eta_0, \qquad \mathscr{Y}_s = N_{C/Y}, \qquad \mathscr{A}_s \xrightarrow[\beta_s]{\sim} T_{Y/B}|_C \times_C N_{C/Y}.$$

Under these identifications, we have the following commutative diagram:

$$\begin{tikzcd}
\Gamma \times \eta_0 \arrow[r, shift left, "t \times \mathrm{id}_{\eta_0}"] \arrow[r, shift right, "s \times \mathrm{id}_{\eta_0}"'] \arrow[d, "j_{\mathscr{A}}"'] & Y \times \eta_0 \arrow[d, "j_{\mathscr{Y}}"] \\
\mathscr{A}_{\mathscr{T}} \arrow[r, shift left, "\widetilde{\mu}_{\mathscr{T}}"] \arrow[r, shift right, "\widetilde{q}_{\mathscr{T}}"'] & \mathscr{Y}_{\mathscr{T}} \\
T_{Y/B}|_C \times_C N_{C/Y} \arrow[r, shift left, "\mu"] \arrow[r, shift right, "q"'] \arrow[u, hook, "i_{\mathscr{A}}"] & N_{C/Y}. \arrow[u, hook, "i_{\mathscr{Y}}"']
\end{tikzcd}$$

The upper arrows give the deformation diagram for $\widetilde{\mu}$, and the lower arrows give that for $\widetilde{q}$. In both cases the generic-fiber and special-fiber squares are cartesian. The generic maps are $t \times \mathrm{id}_{\eta_0}$ and $s \times \mathrm{id}_{\eta_0}$, and the special maps are μ and q of (4.6.21).

Put

$$K_{\eta_0} = \mathscr{H}_G \boxtimes \Lambda_{\eta_0}.$$

Thus

$$\mathscr{K}_G = \Psi_{\mathscr{Y}}(K_{\eta_0}) = i^*_{\mathscr{Y}} Rj_{\mathscr{Y}*} K_{\eta_0}.$$

Here and below Rj_* has the standard oriented-topos interpretation when η_0 is a geometric generic point. The base-change morphism attached to the upper deformation square is the composite

$$\begin{aligned}
\mathrm{BC}_{\widetilde{\mu}} : \quad \mu^* \mathscr{K}_G = \mu^* i^*_{\mathscr{Y}} Rj_{\mathscr{Y}*} K_{\eta_0} &\xrightarrow{\sim} i^*_{\mathscr{A}} \widetilde{\mu}^*_{\mathscr{T}} Rj_{\mathscr{Y}*} K_{\eta_0} \\
&\longrightarrow i^*_{\mathscr{A}} Rj_{\mathscr{A}*} \widetilde{\mu}^*_{\eta_0} K_{\eta_0} = \Psi_{\mathscr{A}}(\widetilde{\mu}^*_{\eta_0} K_{\eta_0}).
\end{aligned}$$

The first arrow is induced by the cartesian special-fiber square, and the second arrow is the base-change transformation

$$\widetilde{\mu}^*_{\mathscr{T}} Rj_{\mathscr{Y}*} \longrightarrow Rj_{\mathscr{A}*} \widetilde{\mu}^*_{\eta_0}.$$

Since $\widetilde{\mu}_{\mathscr{T}}$ is smooth, smooth base change for nearby cycles [Sai26, (1.52)–(1.53)] implies that $\mathrm{BC}_{\widetilde{\mu}}$ is an isomorphism. It is the first isomorphism in (4.6.22).

Similarly, the lower deformation square gives

$$\begin{aligned}
\mathrm{BC}_{\widetilde{q}} : \quad q^* \mathscr{K}_G = q^* i^*_{\mathscr{Y}} Rj_{\mathscr{Y}*} K_{\eta_0} &\xrightarrow{\sim} i^*_{\mathscr{A}} \widetilde{q}^*_{\mathscr{T}} Rj_{\mathscr{Y}*} K_{\eta_0} \\
&\longrightarrow i^*_{\mathscr{A}} Rj_{\mathscr{A}*} \widetilde{q}^*_{\eta_0} K_{\eta_0} = \Psi_{\mathscr{A}}(\widetilde{q}^*_{\eta_0} K_{\eta_0}).
\end{aligned}$$

Since $\widetilde{q}_{\mathscr{T}}$ is smooth, $\mathrm{BC}_{\widetilde{q}}$ is an isomorphism. In the following chain, it is written from right to left.

Under the generic-fiber identifications in the diagram, (4.6.5) pulls back to

$$\epsilon_{\eta_0} : \widetilde{\mu}^*_{\eta_0} K_{\eta_0} \xrightarrow{\sim} \widetilde{q}^*_{\eta_0} K_{\eta_0}.$$

Applying nearby cycles on $\mathscr{A}_{\mathscr{T}}$ and using the two base-change isomorphisms, we obtain

$$\mu^* \mathscr{K}_G \xrightarrow[\sim]{\mathrm{BC}_{\widetilde{\mu}}} \Psi\big(\widetilde{\mu}^*_{\eta_0}(\mathscr{H}_G \boxtimes \Lambda_{\eta_0})\big) \xrightarrow{\Psi(\epsilon_{\eta_0})} \Psi\big(\widetilde{q}^*_{\eta_0}(\mathscr{H}_G \boxtimes \Lambda_{\eta_0})\big) \xleftarrow[\sim]{\mathrm{BC}_{\widetilde{q}}} q^* \mathscr{K}_G. \tag{4.6.22}$$

Notice that the left-pointing arrow is the base-change map $\mathrm{BC}_{\widetilde{q}}$ displayed above. Hence the specialized equivariance isomorphism is

$$\theta = \mathrm{BC}_{\widetilde{q}}^{-1} \circ \Psi(\epsilon_{\eta_0}) \circ \mathrm{BC}_{\widetilde{\mu}} : \mu^* \mathscr{K}_G \xrightarrow{\sim} q^* \mathscr{K}_G.$$

The unit and multiplication diagrams for the deformed groupoid commute, and (4.6.5) is the identity descent datum. By the naturality of smooth base change, we have

$$e^*\theta = \mathrm{id}_{\mathscr{K}_G}$$

and (4.3.3). Thus $\mathscr{K}_G$ is strongly equivariant for the image in $N_{C/Y}$ of the vertical tangent bundle $T_{Y/B}|_C$.

We now identify this image. At $c = (q_1, q_2) = (y, H_1, H_2)$, we have

$$T_{Y/B,c} = T_{Q/P^\vee,q_1} \oplus T_{Q/P^\vee,q_2}.$$

Under the canonical identification $N_{C/Y} \simeq C \times_P TP$, the map in (4.6.21) sends

$$(u_1, u_2) \longmapsto dp(u_1) - dp(u_2). \tag{4.6.23}$$

Indeed, the normal coordinate is the first-order difference between the two P-coordinates. Hence (4.6.23) agrees with α in (4.5.1). This proves the asserted strong equivariance.

The fiber of $h : Q \to P^\vee$ over H_i is H_i, and dp identifies $T_{Q/P^\vee,q_i}$ with T_yH_i. Formula (4.6.23) therefore gives

$$\operatorname{im}\alpha_{q_1,q_2} = T_yH_1 + T_yH_2,$$

which is (4.6.1). Two distinct hyperplanes in T_yP span T_yP, so α is surjective away from Δ_Q. A nonzero covector ξ is annihilated by $\alpha^\vee_{q_1,q_2}$ if and only if it vanishes on both T_yH_1 and T_yH_2. This is equivalent to

$$T_yH_1 = T_yH_2 = \ker\xi.$$

There is a unique hyperplane $H_\xi \subset P$ through y having this tangent space. Hence $q_1 = q_2 = (y, H_\xi)$, which proves (4.6.2). $\square$

4.7. We finish the section by combining ULA constancy with the two Fourier-support lemmas.

Lemma 4.8 (ULA collision constancy)**.** *Let $W \subset Q$ be open and suppose $G|_W$ is ULA over P. Put $C_W = C(W, W)$ and $N_W = N(W, W)$. The universal specialization morphism is an isomorphism*

$$\mathrm{pr}^*_{C_W}(\mathscr{H}_G|_{C_W}) \xrightarrow{\sim} \mathscr{K}_G|_{N_W}. \tag{4.8.1}$$

In particular, the right side is constant in the TP-coordinate.

Proof. Put $p_W = p|_W$ and $L = G|_W$. By hypothesis, L is ULA over P. By [LZ22, Theorem 2.16 and Corollary 2.18], its relative Verdier dual

$$\mathbb{D}_{W/P}L = R\mathcal{H}om(L, K_{W/P})$$

is also ULA over P. Since P is smooth over k, the complex $K_{P/k}$ is invertible up to shift and Tate twist, and transitivity of dualizing complexes gives

$$K_{W/k} \simeq K_{W/P} \otimes p_W^* K_{P/k}.$$

Since L has finite Tor dimension, we have

$$\mathbb{D}_W L \simeq \mathbb{D}_{W/P}L \otimes p_W^* K_{P/k}. \tag{4.8.2}$$

Tensoring with the pullback of an invertible lisse complex preserves ULA. Hence $\mathbb{D}_W L$ is ULA over P.

By smooth purity for the second projection $W \times W \to W$ and the dualizability of L, we obtain

$$\mathscr{H}_G|_{W\times W} \simeq R\mathcal{H}om(\mathrm{pr}_1^* L, \mathrm{pr}_2^! L) \simeq \mathbb{D}_W L \boxtimes L. \tag{4.8.3}$$

Universal local acyclicity is stable under external products. Indeed, by [LZ22, Theorem 2.16], the pairs $(W, \mathbb{D}_W L)$ and (W, L) are dualizable over P, their tensor product is dualizable over $P \times P$, and the converse direction of the same theorem gives ULA. Hence the kernel in (4.8.3) is ULA for

$$f = p_W \times p_W : W \times W \longrightarrow P \times P.$$

Put

$$\mathscr{D}_W = \mathscr{D}_{C_W}(W \times W), \qquad \mathscr{D}_P = \mathscr{D}_{\Delta_P}(P \times P).$$

Let r_W and r_P denote the morphisms from these deformation spaces to $W \times W$ and $P \times P$, respectively. By lemma 3.2, there is a cartesian square

$$\begin{tikzcd} \mathscr{D}_W \arrow[r, "\Pi_W"] \arrow[d, "r_W"'] & \mathscr{D}_P \arrow[d, "r_P"] \\ W \times W \arrow[r, "f"'] & P \times P. \end{tikzcd} \tag{4.8.4}$$

By the base-change stability of ULA,

$$\widetilde{\mathscr{H}_W} := r_W^*(\mathscr{H}_G|_{W \times W})$$

is ULA relative to Π_W.

The morphism

$$\pi_P : \mathscr{D}_P \longrightarrow \mathbf{A}^1$$

is smooth. Indeed, in an étale coordinate chart of P, the deformation of the diagonal has coordinates (y, v, τ) with

$$y' = y + \tau v;$$

and π_P is the projection to τ. A smooth morphism is universally locally acyclic relative to the constant complex. Applying the transitivity of ULA to $\mathscr{D}_W \xrightarrow{\Pi_W} \mathscr{D}_P \xrightarrow{\pi_P} \mathbf{A}^1$, we conclude that $\widetilde{\mathscr{H}_W}$ is ULA over $\mathbf{A}^1$.

We base change to the henselian trait at $0 \in \mathbf{A}^1$ and denote the generic- and special-fiber immersions by j and i. Since the vanishing cycles of a ULA complex vanish, the universal specialization morphism

$$i^*\widetilde{\mathscr{H}_W} \longrightarrow R\Psi(j^*\widetilde{\mathscr{H}_W}) \tag{4.8.5}$$

is an isomorphism. The special fiber of $\mathscr{D}_W$ is

$$N_W = N_{C_W/(W\times W)} \simeq C_W \times_P TP,$$

and the restriction of r_W to N_W is the composite

$$N_W \xrightarrow{\mathrm{pr}_{C_W}} C_W \hookrightarrow W \times W.$$

Hence the source of (4.8.5) is

$$\mathrm{pr}_{C_W}^*(\mathscr{H}_G|_{C_W}).$$

By the definition of Verdier specialization and its compatibility with the open restriction $C_W \subset C$, the target is

$$\nu_{C_W}(\mathscr{H}_G|_{W\times W}) \simeq (\nu_C \mathscr{H}_G)|_{N_W} = \mathscr{K}_G|_{N_W}.$$

Thus (4.8.5) is (4.8.1). Its source is pulled back from C_W along $N_W = C_W \times_P TP \to C_W$, and hence so is its target. This proves the constancy in the TP-coordinate. $\square$

Under the canonical identification $\mathbf{P}(T^*P) \simeq Q$, the Legendre projection is

$$\lambda : T^*P \setminus T^*_P P \longrightarrow Q, \qquad (y, \xi) \longmapsto (y, H_\xi), \qquad T_y H_\xi = \ker \xi. \tag{4.8.6}$$

This is a $\mathbf{G}_m$-torsor. For an open subset $W \subset Q$, put

$$\mathcal{T}^\times_W = \lambda^{-1}(W) \subset T^*P \setminus T^*_P P. \tag{4.8.7}$$

Then $\mathcal{T}^\times_W$ is open.

In the rest of this section, let $F \in D^b_{\mathrm{ctf}}(P, \Lambda)$ and $G = h^*\mathbf{R}(F)[n-1](n-1)$. Recall the notation $\varrho : N = C \times_P TP \to TP$.

Theorem 4.9 (Proper collision Fourier localization)**.** *If $G|_W$ is ULA over P, then*

$$\mathfrak{F}_{\psi,TP/P}\big(R\varrho_*\mathscr{K}_G\big)\big|_{\mathcal{T}^\times_W} = 0. \tag{4.9.1}$$

Proof. Put $U = C(W, W)$ and $Z = C \setminus U$, and denote by

$$a_U : U \longrightarrow P, \qquad a_Z : Z \longrightarrow P$$

the restrictions of $C \to P$. Put

$$N_U = U \times_P TP, \qquad N_Z = Z \times_P TP,$$

and let $j : N_U \hookrightarrow N$ and $i : N_Z \hookrightarrow N$ denote the corresponding open and closed immersions. We have the localization triangle

$$j_!j^*\mathscr{K}_G \longrightarrow \mathscr{K}_G \longrightarrow i_*i^*\mathscr{K}_G \xrightarrow{+1} . \tag{4.9.2}$$

On N_U, lemma 4.8 gives

$$j^*\mathscr{K}_G \simeq \mathrm{pr}_U^*(\mathscr{H}_G|_U). \tag{4.9.3}$$

Let

$$b_U = a_U \times \mathrm{id}_{TP} : N_U \longrightarrow TP;$$

thus $b_U = \varrho j$. The morphism a_U need not be proper, but ϱ is proper since it is the base change of the projective morphism $C \to P$. By properness and the composition law for compactly supported direct image, we have

$$\begin{aligned} R\varrho_*j_!j^*\mathscr{K}_G &\simeq R\varrho_!j_!j^*\mathscr{K}_G \simeq R(\varrho j)_!j^*\mathscr{K}_G \\ &\simeq Rb_{U!}\,\mathrm{pr}_U^*(\mathscr{H}_G|_U). \end{aligned} \tag{4.9.4}$$

Applying lemma 4.2 to $a_U : U \to P$, we obtain

$$\mathfrak{F}_{\psi,TP/P}R\varrho_*j_!j^*\mathscr{K}_G\big|_{T^*P\setminus T^*_P P} = 0. \tag{4.9.5}$$

The morphism a_Z is proper since $C \to P$ is projective and Z is closed in C. Translation through α fixes the C-coordinate and therefore preserves N_Z. Hence the strong $\mathcal{E}$-equivariant structure of lemma 4.6 restricts to a strong $\mathcal{E}|_Z$-equivariant structure on $i^*\mathscr{K}_G$ for

$$\alpha_Z = \alpha|_Z : \mathcal{E}|_Z \longrightarrow TP|_Z.$$

Let

$$b_Z = a_Z \times \mathrm{id}_{TP} : N_Z \longrightarrow TP;$$

then $b_Z = \varrho i$ and

$$R\varrho_*i_*i^*\mathscr{K}_G \simeq Rb_{Z*}i^*\mathscr{K}_G. \tag{4.9.6}$$

Applying lemma 4.4 to a_Z and using lemma 4.6, we obtain

$$\begin{aligned} &\mathrm{supp}\,\mathfrak{F}_{\psi,TP/P}R\varrho_*i_*i^*\mathscr{K}_G \cap (T^*P \setminus T^*_P P) \\ &\quad\subset \left\{ \xi \in T^*P \setminus T^*_P P \;\middle|\; \begin{array}{c} \text{there is } z = (q_1, q_2) \in Z \text{ over } y \text{ such that} \\ \alpha_z^\vee(\xi) = 0 \end{array} \right\} \end{aligned}$$

$$\subset \{\xi \in T^*P \setminus T^*_P P \mid (\lambda(\xi), \lambda(\xi)) \in Z\}\,. \tag{4.9.7}$$

The last inclusion follows from (4.6.2). For a diagonal point, we have

$$(q,q) \in Z = C \setminus C(W,W) \iff q \notin W.$$

Thus the last set in (4.9.7) is $\lambda^{-1}(Q \setminus W)$ and is disjoint from $\mathcal{T}_W^\times = \lambda^{-1}(W)$. Hence

$$\mathfrak{F}_{\psi,TP/P} R\varrho_* i_* i^* \mathscr{K}_G\big|_{\mathcal{T}_W^\times} = 0. \tag{4.9.8}$$

Applying $R\varrho_*$ and the triangulated Fourier–Deligne functor to (4.9.2), and then restricting to $\mathcal{T}_W^\times$, the first and third terms vanish by (4.9.5) and (4.9.8). Therefore the middle term vanishes, which proves (4.9.1). □

Corollary 4.10 (Local microlocal vanishing)**.** *If $G|_W$ is ULA over P, then*

$$\mu\mathcal{H}om(F,F)|_{\mathcal{T}_W^\times} = 0. \tag{4.10.1}$$

Proof. Put

$$A = \nu_{\Delta_P}(\mathscr{H}_F), \qquad B = R\varrho_*\nu_C(\mathscr{H}_G) = R\varrho_*\mathscr{K}_G.$$

The specialized Radon retraction (3.8.4) consists of morphisms

$$A \xrightarrow{U_F^{\mathrm{sp}}} B \xrightarrow{V_F^{\mathrm{sp}}} A, \qquad V_F^{\mathrm{sp}} U_F^{\mathrm{sp}} = \mathrm{id}_A\,.$$

Applying the triangulated Fourier–Deligne functor, which preserves compositions and identity morphisms, we obtain a retraction

$$\mu\mathcal{H}om(F,F) \xrightarrow{\mathfrak{F}_{\psi,TP/P}(U_F^{\mathrm{sp}})} \mathfrak{F}_{\psi,TP/P} R\varrho_*\mathscr{K}_G \xrightarrow{\mathfrak{F}_{\psi,TP/P}(V_F^{\mathrm{sp}})} \mu\mathcal{H}om(F,F). \tag{4.10.2}$$

Here we use the defining identification

$$\mu\mathcal{H}om(F,F) = \mathfrak{F}_{\psi,TP/P}\nu_{\Delta_P}(\mathscr{H}_F).$$

Let

$$\iota_W : \mathcal{T}_W^\times \hookrightarrow T^*P$$

be the open immersion. Applying ι_W^* to (4.10.2), we again obtain a retraction. By theorem 4.9, its middle term is zero. Hence the identity of $\iota_W^*\mu\mathcal{H}om(F,F)$ factors through the zero object and is the zero morphism. An object in an additive derived category with zero identity morphism is zero. Therefore

$$\iota_W^*\mu\mathcal{H}om(F,F) = 0,$$

which proves (4.10.1). □

5. Proof of the microlocal inclusion

5.1. We first treat the change of coefficients. In general, the singular support of a rational ℓ-adic sheaf cannot be calculated from an arbitrary integral model. We therefore make the microlocal construction over E in the pro-étale category. Finite reductions will only be used to verify the canonical identities.

Lemma 5.2 (Pro-étale realization of the microlocal construction)**.** *Let $E/\mathbf{Q}_\ell$ be finite, let $\mathcal{O} = \mathcal{O}_E$, and let π be a uniformizer.*

(i) *For every finite-type k-scheme Y, the constructible pro-étale category with $\mathcal{O}$-coefficients is derived π-complete and is recovered from the categories with coefficients $\mathcal{O}/\pi^m$. Every object of the constructible E-category has an integral model.*

(ii) *The operations used to form the diagonal endomorphism kernel, Verdier specialization, and Fourier transform preserve constructibility and are compatible with reduction. Hence, for an integral model F_0 of $F \in D_{\mathrm{cons}}(Y_{\mathrm{pro\acute{e}t}}, E)$ and $F_m = F_0 \otimes^L_{\mathcal{O}} \mathcal{O}/\pi^m$, there are canonical isomorphisms*

$$\mu\mathcal{H}om_{\mathcal{O}}(F_0, F_0) \otimes^L_{\mathcal{O}} \mathcal{O}/\pi^m \xrightarrow{\sim} \mu\mathcal{H}om_{\mathcal{O}/\pi^m}(F_m, F_m), \tag{5.2.1}$$

$$\mu\mathcal{H}om_E(F, F) \xrightarrow{\sim} \mu\mathcal{H}om_{\mathcal{O}}(F_0, F_0) \otimes_{\mathcal{O}} E. \tag{5.2.2}$$

The right side of (5.2.2) is independent of F_0 because it realizes the intrinsically defined object on $Y_{\mathrm{pro\acute{e}t}}$.

(iii) *Universal local acyclicity in the pro-étale category is equivalent to dualizability in the relative correspondence category. It is preserved by smooth pullback and proper pushforward. For integral coefficients it can be checked after reduction modulo π, and it can be checked after a faithfully flat extension of the coefficient field. Moreover, Beilinson's singular support and its projective Radon description hold for constructible E-sheaves.*

Proof. By the constructible pro-étale comparison of Hemo–Richarz–Scholbach, there is an equivalence of stable ∞-categories

$$D_{\mathrm{cons}}(Y_{\mathrm{pro\acute{e}t}}, \mathcal{O}) \simeq \varprojlim_m D^b_{\mathrm{ctf}}(Y, \mathcal{O}/\pi^m).$$

Rationalization and the existence of integral models follow from Hansen–Scholze; see [Bar24, §2.1]. By [Bar24, Lemma 2.1], reduction modulo π is conservative on derived complete constructible objects. This proves (i).

The diagonal kernel is the internal Hom of perfect constructible objects and hence commutes with perfect change of coefficients. Verdier specialization is defined by nearby cycles on the deformation space. Its constructibility and its compatibility with finite reduction are given in [Bar24, §2.3]. Moreover, compactly supported direct image, tensor product, proper direct image, and the Künneth and projection-formula morphisms define symmetric monoidal coefficient-change functors on the relative correspondence categories [Bar24, §3.1]. Applying these compatibilities to the diagonal kernel, Verdier specialization, and Fourier transformation, successively, we obtain (5.2.1). By derived completeness,

$$\mu\mathcal{H}om_{\mathcal{O}}(F_0, F_0) \simeq R\varprojlim_m \mu\mathcal{H}om_{\mathcal{O}/\pi^m}(F_m, F_m),$$

and rationalization gives (5.2.2). This proves (ii).

The equivalence between ULA and relative dualizability follows from [Bar24, Proposition 3.6]. The required stability is [Bar24, Lemma 3.6], and the assertions on reduction and extension of scalars are [Bar24, Lemmas 3.10 and 3.11]. Finally, the existence and the formal properties of ℓ-adic singular support follow from [Bar24, Theorem 1.5], and the projective Radon description follows from [Bar24, Proposition 4.5]. This proves (iii). $\square$

5.3. We now prove the main inclusion. We first treat finite coefficients and then pass to the constructible pro-étale category.

Theorem 5.4. *For every smooth k-scheme X and every $F \in D^b_{\mathrm{ctf}}(X, \Lambda)$, we have*

$$\mathrm{SS}_\mu(F) \subseteq \mathrm{SS}(F). \tag{5.4.1}$$

The same assertion holds for constructible complexes with coefficients in a finite extension $E/\mathbf{Q}_\ell$ on the pro-étale site.

Proof. Suppose first that $X = P$. Let $q \in Q \setminus E_F^{\mathrm{Rad}}$, and take an open neighborhood $W \subset Q \setminus E_F^{\mathrm{Rad}}$ of q. Put $G = h^*\mathbf{R}(F)[d](d)$. Then $G|_W$ is ULA over P. By corollary 4.10, we have

$$\mu\mathcal{H}om(F, F)|_{\mathcal{T}_W^\times} = 0.$$

Since q is arbitrary, this proves (2.3.6). Now proposition 2.4, including the zero-section argument, gives (5.4.1) for every smooth X.

Let now F have coefficients in a finite extension $E/\mathbf{Q}_\ell$. We work in $D_{\mathrm{cons}}(-_{\text{proét}}, E)$. The arrows in proposition 3.5 and theorem 3.8 are obtained from the unit and counit of adjunction, proper base change, the projection formula, relative duality, and nearby cycles. By lemma 5.2, these transformations exist in the pro-étale category, are compatible with composition, and preserve the triangle identities. Hence the proper collision comparison and the retraction hold over E. Notice that no ULA lattice is chosen.

Let q be outside the E-adic Radon non-ULA locus. By Barrett's Radon criterion, there is an open neighborhood W of q such that $G|_W$ is ULA over P. By [Bar24, Proposition 3.6], it is relatively dualizable. The proof of lemma 4.8 therefore shows that the open collision term is constant in the tangent variable. The strong equivariance of the closed term is formulated in the same correspondence category and remains valid. The two Fourier orthogonality lemmas only use the equality

$$R\Gamma_c(\mathbf{A}^1_{\bar{k}}, \mathcal{L}_\psi(cx)) = 0 \qquad (c \neq 0),$$

which is preserved under extension of scalars from a finite coefficient ring containing the values of ψ. Thus theorem 4.9 and corollary 4.10 give (2.3.6) over E. The ℓ-adic Radon description [Bar24, Proposition 4.5] and proposition 2.4 prove the assertion. The case of an algebraic coefficient extension follows by approximation from [Bar24, Theorem 1.5(iii) and Lemma 3.11]. □

Remark 5.5 (Properness)**.** The projectivity of the incidence projections is used for the Radon adjunction, proper duality, and the trace calculation in lemma 3.7. The properness supplied by the full collision center has two further consequences. First, nearby cycles commute with $R\Pi_*$ in proposition 3.5. Second, ϱ is proper; hence the image of the closed equivariant support in lemma 4.4 is closed, and the open term can be written using compact support. Thus no compactification of a contact chart and no boundary transversality or tame-boundary condition is required.

5.6. Let $i : Z \hookrightarrow X$ be a closed immersion of smooth k-schemes. Put

$$N = N_{Z/X}, \qquad E = T_X|_Z,$$

and write

$$0 \longrightarrow T_Z \longrightarrow E \xrightarrow{q} N \longrightarrow 0, \qquad q^\vee : N^\vee = T^*_Z X \hookrightarrow E^\vee = T^*X|_Z. \tag{5.6.1}$$

Saito's microlocalization of F along Z is

$$\mu_Z(F) = \mathfrak{F}_{\psi, N/Z}(\nu_Z F) \quad \text{on } N^\vee \times_k \eta_0. \tag{5.6.2}$$

As before, we suppress the factor η_0 in support statements. We first construct an action on the deformation space. Let $s, t : X \times X \rightrightarrows X$ be the source and target of the pair groupoid, with $s = \mathrm{pr}_1$ and $t = \mathrm{pr}_2$.

Lemma 5.7 (Deformed pair-groupoid action)**.** *The deformation $\mathscr{D}_{\Delta_X}(X \times X)$ is a groupoid over $X \times \mathbf{A}^1$ and acts canonically on $\mathscr{D}_Z(X)$. On the generic fiber the action is the pair-groupoid action. On the special fiber it is*

$$a_0 : E \times_Z N \longrightarrow N, \qquad (v, \bar{w}) \longmapsto q(v) + \bar{w}. \tag{5.7.1}$$

The zero section of E is the unit, and (5.7.1) is associative for addition in E.

Proof. Write

$$G = X \times X, \qquad \mathscr{G} = \mathscr{D}_{\Delta_X}(G), \qquad \mathscr{X} = \mathscr{D}_Z(X), \qquad \mathscr{B} = \mathscr{D}_X(X) = X \times \mathbf{A}^1.$$

The morphisms $s, t : (G, \Delta_X) \to (X, X)$ are morphisms of pairs. By the functoriality of the extended Rees algebra, they induce morphisms

$$\widetilde{s}, \widetilde{t} : \mathscr{G} \rightrightarrows \mathscr{B}.$$

The unit $e : X \to G$, the multiplication $m : G \times_{t,X,s} G \to G$, and the pair-groupoid action $a : G \times_{s,X} X \to X$ preserve the corresponding centers. Moreover, deformation commutes with the two fiber products in question. Namely, there are canonical isomorphisms

$$\begin{aligned}\mathscr{D}_{\Delta_X \times_X \Delta_X}(G \times_{t,X,s} G) &\simeq \mathscr{G} \times_{\widetilde{t},\mathscr{B},\widetilde{s}} \mathscr{G},\\ \mathscr{D}_Z(G \times_{s,X} X) &\simeq \mathscr{G} \times_{\widetilde{s},\mathscr{B}} \mathscr{X}.\end{aligned}$$

Indeed, both sides are represented by the same extended Rees algebra. In the coordinates below, this means that the differences of two consecutive points are divisible by τ. The deformations of e, m, and a therefore define a groupoid structure on $\mathscr{G}$ over $\mathscr{B}$ and an action

$$\widetilde{a} : \mathscr{G} \times_{\widetilde{s},\mathscr{B}} \mathscr{X} \longrightarrow \mathscr{X}. \tag{5.7.2}$$

The groupoid and action identities follow by applying the functorial Rees construction to the corresponding identities before deformation.

It remains to calculate the special fiber of (5.7.2). Étale locally on X, take coordinates $x_1, \dots, x_n$ such that Z is defined by $x_1 = \cdots = x_c = 0$. Write (x, y) for an arrow with source x and target y. On $\mathscr{X}$, put

$$x_i = \tau w_i \quad (1 \le i \le c),$$

and, on $\mathscr{G}$, put

$$y_i - x_i = \tau v_i \quad (1 \le i \le n).$$

Then the fiber product in (5.7.2) has coordinates $(x_{c+1}, \dots, x_n, w_1, \dots, w_c, v_1, \dots, v_n, \tau)$. For $1 \le i \le c$, we have

$$y_i = x_i + \tau v_i = \tau(w_i + v_i),$$

whereas, for $i > c$, we have $y_i = x_i + \tau v_i$, whose restriction to the special fiber is x_i. Hence the special-fiber action is

$$(v_1, \dots, v_n; w_1, \dots, w_c) \longmapsto (w_1 + v_1, \dots, w_c + v_c).$$

This is the map $(v, \bar{w}) \mapsto q(v) + \bar{w}$. Since the quotient map q in (5.6.1) is intrinsic, the local maps agree on overlaps. The special fiber of the deformed unit is the zero section $0_E : Z \to E$. Moreover,

$$a_0(0, \bar{w}) = \bar{w}, \qquad a_0(v_2, a_0(v_1, \bar{w})) = \bar{w} + q(v_1 + v_2),$$

which proves the unit and associativity identities. □

Put

$$\mathscr{H}_F = R\mathcal{H}om(s^*F, t^!F), \qquad A_F = \nu_{\Delta_X}(\mathscr{H}_F), \qquad M_{Z,F} = \nu_Z F,$$

and let $A_{F,Z}$ be the restriction of A_F to $E = T_X|_Z$. For complexes A on E and M on N, define the action convolution

$$A \star M = Ra_{0!}\big(\mathrm{pr}_1^* A \otimes \mathrm{pr}_2^* M\big). \tag{5.7.3}$$

Proposition 5.8 (Unital microlocal module)**.** *There is a canonical action*

$$A_{F,Z} \star M_{Z,F} \longrightarrow M_{Z,F} \tag{5.8.1}$$

whose unit is the specialization of id_F*. After Fourier transformation, it gives*

$$(q^\vee)^* \mu\mathcal{H}om(F, F) \otimes \mu_Z(F) \longrightarrow \mu_Z(F). \tag{5.8.2}$$

This action is unital: the composite

$$\mu_Z(F) \simeq \Lambda_{N^\vee} \otimes \mu_Z(F) \longrightarrow (q^\vee)^* \mu\mathcal{H}om(F, F) \otimes \mu_Z(F) \longrightarrow \mu_Z(F) \tag{5.8.3}$$

is the identity.

Proof. We first construct the action on the generic fiber. Put

$$P_\eta = (X \times X) \times_{s,X} X.$$

Let $p_G : P_\eta \to X \times X$ and $p_X : P_\eta \to X$ be the projections, and let $a_\eta : P_\eta \to X$ be the action. Under the canonical isomorphism $P_\eta \simeq X \times X$, we have $p_X = s$ and $a_\eta = t$. Pulling back the evaluation morphism, we obtain

$$p_X^* F \otimes p_G^* \mathscr{H}_F = s^* F \otimes R\mathcal{H}om(s^* F, t^! F) \longrightarrow a_\eta^! F. \tag{5.8.4}$$

By the adjunction $Ra_{\eta!} \dashv a_\eta^!$, it defines a morphism

$$\alpha_\eta : Ra_{\eta!}\big(p_G^* \mathscr{H}_F \otimes p_X^* F\big) \longrightarrow F. \tag{5.8.5}$$

Under adjunction, the identity endomorphism of F defines a cohomological correspondence

$$u_\eta : \Delta_{X!}\Lambda_X \longrightarrow \mathscr{H}_F. \tag{5.8.6}$$

The restriction of the action map to $\Delta_X \times_{s,X} X \simeq X$ is the identity. Therefore the composite

$$F \simeq Ra_{\eta!}\big(p_G^* \Delta_{X!}\Lambda_X \otimes p_X^* F\big) \xrightarrow{u_\eta} Ra_{\eta!}\big(p_G^* \mathscr{H}_F \otimes p_X^* F\big) \xrightarrow{\alpha_\eta} F \tag{5.8.7}$$

is id_F. The composition of cohomological correspondences on the pair groupoid makes $\mathscr{H}_F$ an algebra object, and (5.8.5) defines its action on F. The associativity follows from the associativity of the composition of endomorphisms. In the following, we only use the unit identity (5.8.7).

We next specialize the action. With the notation in the proof of lemma 5.7, put

$$\mathscr{P} = \mathscr{G} \times_{\widetilde{s},\mathscr{B}} \mathscr{X}.$$

The generic and special fibers of $\mathscr{P}$ are $P_\eta \times \mathbf{G}_m$ and $E \times_Z N$, respectively. The special fiber of $\widetilde{a}$ is a_0. Let $\Psi_{\mathscr{G}}$, $\Psi_{\mathscr{X}}$, and $\Psi_{\mathscr{P}}$ denote the nearby-cycle functors on the three deformation spaces. By the base-change morphisms for the two projections from $\mathscr{P}$ and the right-lax tensor morphism for nearby cycles, we obtain a canonical morphism

$$\mathrm{pr}_1^* A_{F,Z} \otimes \mathrm{pr}_2^* M_{Z,F} \longrightarrow \Psi_{\mathscr{P}}\big(p_G^* \mathscr{H}_F \otimes p_X^* F\big). \tag{5.8.8}$$

For a separated morphism of deformation spaces $f : \mathscr{Y} \to \mathscr{W}$, there is an exchange morphism

$$Rf_{0!}\Psi_{\mathscr{Y}}(K) \longrightarrow \Psi_{\mathscr{W}}(Rf_{\eta!}K). \tag{5.8.9}$$

Indeed, take a compactification $\mathscr{Y} \xrightarrow{j} \overline{\mathscr{Y}} \xrightarrow{\bar{f}} \mathscr{W}$. Combine the canonical map $j_{0!}\Psi_{\mathscr{Y}}K \to \Psi_{\overline{\mathscr{Y}}}(j_{\eta!}K)$ with proper base change for $\bar{f}$. This gives (5.8.9), and the construction is independent of the compactification. Notice that (5.8.9) is only a morphism when f is nonproper; it does not assert that nearby cycles commute with $Rf_!$.

Apply $Ra_{0!}$ to (5.8.8). By (5.8.9) and the nearby cycles of (5.8.5), we obtain the composite

$$\begin{aligned} A_{F,Z} \star M_{Z,F} &\longrightarrow Ra_{0!}\Psi_{\mathscr{P}}(p_G^* \mathscr{H}_F \otimes p_X^* F) \\ &\longrightarrow \Psi_{\mathscr{X}} Ra_{\eta!}(p_G^* \mathscr{H}_F \otimes p_X^* F) \xrightarrow{\Psi(\alpha_\eta)} M_{Z,F}. \end{aligned} \tag{5.8.10}$$

This defines (5.8.1). The right-lax symmetric monoidal structure and the exchange morphisms are those for cohomological correspondences constructed in [LZ22, Construction 3.3]. In particular, they are compatible with horizontal composition.

We prove that the action is unital. The deformation of the diagonal unit section is the unit section of $\mathscr{G}$. Since the unit section is closed, proper base change identifies the specialization of (5.8.6) with a morphism

$$u_0 : 0_{E!}\Lambda_Z \longrightarrow A_{F,Z}. \tag{5.8.11}$$

Here we use the canonical isomorphism $\nu_{\Delta_X}(\Delta_{X!}\Lambda_X)|_E \simeq 0_{E!}\Lambda_Z$. The restriction of a_0 to $0_E(Z) \times_Z N$ is the identity of N, so we have a canonical isomorphism

$$(0_{E!}\Lambda_Z) \star M_{Z,F} \xrightarrow{\sim} M_{Z,F}. \tag{5.8.12}$$

By the naturality of the base-change, tensor, and shriek-exchange morphisms, the following diagram is commutative:

$$
\begin{tikzcd}
(0_{E!}\Lambda_Z)\star M_{Z,F} \arrow[r, "u_0\star\mathrm{id}"] \arrow[d, "\sim"'] & A_{F,Z}\star M_{Z,F} \arrow[d] \\
M_{Z,F} \arrow[r, "\mathrm{id}"'] & M_{Z,F}.
\end{tikzcd}
\tag{5.8.13}
$$

Before specialization, this is the unit identity (5.8.7). The compatibility of the right-lax nearby-cycle formalism with units and horizontal composition gives (5.8.13). It follows that

$$M_{Z,F} \simeq (0_{E!}\Lambda_Z)\star M_{Z,F} \longrightarrow A_{F,Z}\star M_{Z,F} \longrightarrow M_{Z,F} \tag{5.8.14}$$

is the identity of $M_{Z,F}$.

We apply the Fourier transformation. Let $A \in D^b_{\mathrm{ctf}}(E,\Lambda)$ and $M \in D^b_{\mathrm{ctf}}(N,\Lambda)$. We first establish the Fourier–convolution isomorphism. Let $\bar\xi \in N^\vee$ be a geometric point over $\bar z \to Z$. By (2.1.3), base change for compactly supported direct image, and the Fubini isomorphism, we have

$$
\begin{aligned}
\big(\mathfrak{F}_{\psi,N/Z}(A\star M)\big)_{\bar\xi} &\simeq R\Gamma_c\big(E_{\bar z}\times N_{\bar z}, A\boxtimes M\otimes \mathcal{L}_\psi(\langle q(v)+w,\bar\xi\rangle)\big)\\
&\simeq R\Gamma_c\big(E_{\bar z}, A\otimes \mathcal{L}_\psi(\langle v, q^\vee(\bar\xi)\rangle)\big)\otimes^L_\Lambda R\Gamma_c\big(N_{\bar z}, M\otimes\mathcal{L}_\psi(\langle w,\bar\xi\rangle)\big)\,.
\end{aligned}
$$

The second isomorphism follows from

$$\langle q(v)+w,\bar\xi\rangle = \langle v, q^\vee(\bar\xi)\rangle + \langle w,\bar\xi\rangle$$

and the Künneth formula. Since these fiberwise isomorphisms are induced by the projection formula and base change for $R(-)_!$, they are functorial in $\bar\xi$. Hence they define a canonical isomorphism

$$\mathfrak{F}_{\psi,N/Z}(A\star M) \xrightarrow{\sim} (q^\vee)^*\mathfrak{F}_{\psi,E/Z}(A)\otimes \mathfrak{F}_{\psi,N/Z}(M). \tag{5.8.15}$$

There is no shift in (5.8.15), since we use the unshifted Fourier transformation of (2.1.3).

Let $j_E : E^\vee = T^*X|_Z \to T^*X$ be the base change of $i : Z \to X$. By base change for the cartesian squares of $T_X|_Z \to T_X$ and of the dual bundles, we have

$$\mathfrak{F}_{\psi,E/Z}(A_{F,Z}) \xrightarrow{\sim} j_E^*\mu\mathcal{H}om(F,F). \tag{5.8.16}$$

Combining (5.8.10) with (5.8.15) and (5.8.16), we obtain (5.8.2). Directly from (2.1.3), the pairing vanishes on the zero section. Therefore

$$\mathfrak{F}_{\psi,E/Z}(0_{E!}\Lambda_Z) \xrightarrow{\sim} \Lambda_{E^\vee}; \tag{5.8.17}$$

see also [Sai26, Lemma 1.1.2(2)]. Applying Fourier transformation to (5.8.13) and using (5.8.15) and (5.8.17), the bottom morphism remains the identity and the upper row becomes the composite in (5.8.3). Thus the action (5.8.2) is unital. $\square$

Corollary 5.9 (Saito's specialization question)**.** *For every smooth closed immersion $Z \hookrightarrow X$ and every $F \in D^b_{\mathrm{ctf}}(X,\Lambda)$,*

$$\operatorname{supp}\mu_Z(F) \subset \mathrm{SS}_\mu(F)\cap T^*_Z X \subset \mathrm{SS}(F)\cap T^*_Z X. \tag{5.9.1}$$

The same assertion holds for constructible complexes with coefficients in a finite extension of $\mathbf{Q}_\ell$. In particular, this answers [Sai26, Question 2.5.3] *affirmatively.*

Proof. Let $\bar\xi$ be a geometric point of $N^\vee = T^*_Z X$ whose image under $q^\vee : N^\vee \hookrightarrow T^*X|_Z \to T^*X$ is not in $\mathrm{SS}_\mu(F)$. We continue to suppress the geometric generic-point factor. By the definition of $\mathrm{SS}_\mu(F)$, we have

$$\big((q^\vee)^*\mu\mathcal{H}om(F,F)\big)_{\bar\xi} = 0.$$

Put $K = \mu_Z(F)_{\bar\xi}$. Taking the stalk at $\bar\xi$ in (5.8.3), the unit identity becomes the factorization

$$K \longrightarrow \big((q^\vee)^*\mu\mathcal{H}om(F,F)\big)_{\bar\xi}\otimes^L_\Lambda K \longrightarrow K$$

of id_K. Since the middle term is zero, we have $\mathrm{id}_K = 0$, and hence $K = 0$. This proves

$$\operatorname{supp} \mu_Z(F) \subset \mathrm{SS}_\mu(F) \cap T^*_Z X.$$

By Theorem 1.4, we have $\mathrm{SS}_\mu(F) \subset \mathrm{SS}(F)$. Intersecting both sides with $T^*_Z X$, we obtain the second inclusion in (5.9.1).

Let the coefficients be in a finite extension $E/\mathbf{Q}_\ell$. All the objects and morphisms in (5.8.10) are defined in the constructible pro-étale correspondence category. The nearby-cycle and six-functor operations satisfy the same right-lax compatibilities. Hence the preceding argument applies. Alternatively, take an integral model $F_{\mathcal{O}_E}$ and put $F_m = F_{\mathcal{O}_E} \otimes^L_{\mathcal{O}_E} \mathcal{O}_E/\pi^m$. The unit diagram for $F_{\mathcal{O}_E}$ reduces, for every m, to (5.8.13) for F_m. By the equivalence

$$D_{\mathrm{cons}}(X_{\text{proét}}, \mathcal{O}_E) \simeq \varprojlim_m D^b_{\mathrm{ctf}}(X, \mathcal{O}_E/\pi^m)$$

and lemma 5.2, the inverse-limit unit morphism is the identity. After inverting π, we obtain (5.8.3) over E, and the same stalk factorization proves (5.9.1). □

6. Non-acyclicity class and equality for perverse sheaves

6.1. In this section, we apply the non-acyclicity class defined in [YZ25]. We first compare its proper push-forward with that of the microlocal characteristic class. We then use a component-detecting pencil to recover each irreducible component of the singular support. The comparison with the non-acyclicity class is made with coefficients in the finite ring Λ. For a finite extension $E/\mathbf{Q}_\ell$, the equality for perverse sheaves follows from the adic microlocal conductor formula and Theorem 1.4. We do not use an adic non-acyclicity class.

Let K be a constructible complex on a smooth k-scheme Y. The identity of K defines Saito's microlocal characteristic class

$$\mathrm{CC}_\mu(K) \in H^0_{\mathrm{SS}_\mu(K)}(T^*Y, e_Y^{\vee *} K_{Y/k}). \tag{6.1.1}$$

Let $f : Y \to S$ be a morphism to a smooth curve, let $s \in S$, and let t be a local parameter at s. After shrinking S around s, we may assume that dt is nowhere zero. Put

$$dt : S \longrightarrow T^*S, \qquad \theta_{f,t} : Y \longrightarrow T^*Y, \quad y \longmapsto d(t \circ f)_y.$$

We impose the following microlocal support condition:

$$\theta_{f,t}^{-1} \mathrm{SS}_\mu(K) \subset Y_s. \tag{6.1.2}$$

Notice that ULA over $S \setminus \{s\}$ does not by itself imply (6.1.2); this is a condition on the cotangent section $\theta_{f,t}$. It will be verified below by Saito's description of the universal characteristic locus. After base change to the henselization $S^h_{(s)}$, condition (6.1.2) defines the pullback with supports

$$I_{f,s}(\mathrm{CC}_\mu(K)) = \theta^*_{f,t} \mathrm{CC}_\mu(K) \in H^0_{Y_s}(Y, K_{Y/k}). \tag{6.1.3}$$

Here the target is the absolute dualizing complex $K_{Y/k}$.

Let $\widetilde{C}^{Y_s}_{Y/S/k}(K)$ denote the non-acyclicity class in $H^0_{Y_s}(Y, K_{Y/S/k})$, where $K_{Y/S/k}$ is the object defined by the distinguished triangle

$$K_{Y/S} \longrightarrow K_{Y/k} \longrightarrow K_{Y/S/k} \xrightarrow{+1}.$$

Thus $I_{f,s}(\mathrm{CC}_\mu(K))$ and $\widetilde{C}^{Y_s}_{Y/S/k}(K)$ lie in different supported cohomology groups. The following proposition compares their proper push-forwards.

Proposition 6.2 (Proper NA–microlocal trace comparison)**.** *Let $f : Y \to S$ be proper, where Y is smooth and S is a smooth connected curve. Suppose that f is ULA relative to $K \in D^b_{\mathrm{ctf}}(Y, \Lambda)$ over $S \setminus \{s\}$ and that (6.1.2) holds. Then*

$$f_* \widetilde{C}^{Y_s}_{Y/S/k}(K) = f_* I_{f,s}(\mathrm{CC}_\mu(K)) = -a_s(Rf_* K)[s] \tag{6.2.1}$$

in $H^0_s(S, K_{S/k})$*, where* a_s *is the Artin conductor.*

Proof. We work over $S^h_{(s)}$ and take cohomology with support on the closed fiber. The cotangent correspondence

$$T^*Y \xleftarrow{df} Y \times_S T^*S \xrightarrow{\mathrm{pr}} T^*S$$

and refined Gysin base change give a commutative diagram

$$\begin{tikzcd} H^0_{\mathrm{SS}_\mu(K)}(T^*Y, e_Y^{\vee *} K_{Y/k}) \arrow[r, "f_!"] \arrow[d, "I_{f,s}"'] & H^0_{C_S}(T^*S, e_S^{\vee *} K_{S/k}) \arrow[d, "I_{dt,s}"] \\ H^0_{Y_s}(Y, K_{Y/k}) \arrow[r, "f_*"'] & H^0_s(S, K_{S/k}), \end{tikzcd}$$

where $C_S = f_\circ \mathrm{SS}_\mu(K) \cup \mathrm{SS}_\mu(Rf_*K)$, and the upper horizontal morphism is taken after change to this support. The diagram is commutative by the projection formula for the regular section dt. Indeed, both composites are the trace of the derived intersection of the cotangent correspondence with dt. By Saito's proper-pushforward formula [Sai26, Proposition 2.3.3], we have

$$f_!\,\mathrm{CC}_\mu(K) = \mathrm{CC}_\mu(Rf_*K).$$

On the curve S, this class is the cycle class of $\mathrm{CC}(Rf_*K)$ by [Sai26, Proposition 2.5.1]. The coefficient of T^*_sS in $\mathrm{CC}(Rf_*K)$ is $-a_s(Rf_*K)$. Since dt is disjoint from the zero section and meets T^*_sS transversally with multiplicity one, it follows that

$$f_*I_{f,s}(\mathrm{CC}_\mu(K)) = -a_s(Rf_*K)[s].$$

By [YZ25, Proposition 4.5 and Theorem 6.5], we have

$$f_*\widetilde{C}^{Y_s}_{Y/S/k}(K) = -a_s(Rf_*K)[s],$$

under the canonical identification of the supported target on S. This proves (6.2.1). Notice that (6.2.1) compares the two classes only after proper push-forward. □

Corollary 6.3 (NA obstruction at a unique characteristic point)**.** *In the situation of proposition 6.2, assume that* $z \in Y_s$ *is the only non-ULA point over* s *and that the residue degree* $[k(z) : k(s)]$ *is invertible in* Λ*. If* $\theta_{f,t}$ *does not meet* $\mathrm{SS}_\mu(K)$ *over* Y_s*, then the local class* $\widetilde{C}^{\{z\}}(K, f)$ *is zero. If, moreover, the vanishing cycles are concentrated in one degree and their total dimension has nonzero image in* Λ*, this contradicts the cohomological Milnor formula.*

Proof. Since $\theta_{f,t}$ does not meet $\mathrm{SS}_\mu(K)$ over Y_s, we have $I_{f,s}(\mathrm{CC}_\mu(K)) = 0$. By proposition 6.2, the proper push-forward of the NA class is zero. By excision, this is the push-forward of the class supported at z. The trace from z to s is multiplication by $[k(z) : k(s)]$, which is a unit in Λ. Hence $\widetilde{C}^{\{z\}}(K, f) = 0$.

On the other hand, by the cohomological Milnor formula [YZ25, Theorem 6.2] identifies this class with the negative alternating total dimension of $R\Phi_z(K, f)$. If the vanishing cycles are concentrated in one degree and their total dimension is nonzero in Λ, the class is nonzero. This is a contradiction. □

Proposition 6.4 (Adic microlocal conductor identity)**.** *Let* $E/\mathbf{Q}_\ell$ *be finite. Under the geometric hypotheses of proposition 6.2, for* $K \in D^b_{\mathrm{ctf}}(Y, E)$ *one has*

$$f_*I_{f,s}(\mathrm{CC}_\mu(K)) = -a_s(Rf_*K)[s] \quad \textit{in } H^0_s(S, K_{S/k}) \otimes E. \tag{6.4.1}$$

Proof. We work in $D_{\mathrm{cons}}(Y_{\text{proét}}, E)$. The identity of K, the evaluation morphism, specialization, and Fourier transformation define $\mathrm{CC}_\mu(K)$ in this category. After choosing an integral model, lemma 5.2 identifies it with the rationalization of the compatible finite-level classes.

We first prove the proper-pushforward identity over E. The two composites in Saito's proper-pushforward diagram are obtained from proper base change, the projection formula, specialization, and evaluation. After multiplication by a power of π, the objects and both composites admit integral models. For every m, their reductions modulo π^m agree by [Sai26, Proposition 2.3.3]. Derived π-completeness, or the conservativity of reduction [Bar24, Lemma 2.1], shows that the two integral morphisms agree. After rationalization, we obtain

$$f_! \operatorname{CC}_\mu(K) = \operatorname{CC}_\mu(Rf_*K).$$

The same argument shows that the refined Gysin diagram in the proof of proposition 6.2 is commutative in the pro-étale E-category. On the curve S, the finite-level calculation [Sai26, Proposition 2.5.1] is compatible with coefficient change. It identifies the coefficient of T^*_sS with the integral alternating Artin conductor. Since dt meets T^*_sS with multiplicity one, we obtain (6.4.1). In particular, the proof does not calculate singular support from an arbitrary reduction. □

6.5. Let $i : X \to \mathbb{P} = \mathbf{P}(W^\vee)$ be an immersion, and let $C_a \subset \operatorname{SS}(\mathcal{P})$ be an irreducible component. Let $\widetilde{C}_a \subset X \times_{\mathbb{P}} T^*\mathbb{P}$ be its inverse image. For a line $L \subset \mathbb{P}^\vee$, let A_L denote its axis. Put

$$Q_L = \{(u, H') \in \mathbb{P} \times L \mid u \in H'\}, \qquad p_L : Q_L \to \mathbb{P}, \qquad g_L : Q_L \to L.$$

Then Q_L is the blow-up of $\mathbb{P}$ along A_L, and g_L is proper.

Lemma 6.6 (Saito's component-detecting pencil, [Sai17, Lemma 3.10])**.** *Let $\mathcal{P}$ be a perverse E-sheaf on X. After a Veronese embedding of degree at least 3, every component $C_a \subset \operatorname{SS}(\mathcal{P})$ has a dense open set in the projectivization of its lift $\mathbf{P}(\widetilde{C}_a)$ with the following property. For a point of this open represented by (x, H) and for a general line L through H, one has $x \notin A_L$; after replacing L by an open neighborhood $V \subset L$ of H, set*

$$G_L = p_L^* i_! \mathcal{P}|_{g_L^{-1}(V)}, \qquad g = g_L|_{g_L^{-1}(V)}, \qquad z = (x, H).$$

Then the following assertions hold:

(i) *g is ULA relative to G_L away from the single point z over H;*

(ii) *$R^q\Phi_z(G_L, g) = 0$ for $q \neq -1$, while $R^{-1}\Phi_z(G_L, g) \neq 0$;*

(iii) *$d(g|_X)_x$ is the chosen covector of C_a.*

Proof. We give an adic proof. After the Veronese embedding, Barrett's ℓ-adic form of Beilinson's Radon theorem associates to C_a a unique irreducible component D_a of the ramification divisor of $\mathbf{R}(i_!\mathcal{P})$. The generic morphism

$$\mathbf{P}(\widetilde{C}_a) \longrightarrow D_a$$

is radicial by [Bar24, Proposition 4.6]. Take a dense open subset $D_a^\circ \subset D_a$, disjoint from the other discriminant components, over which this morphism is finite and radicial. For every geometric point $H \in D_a^\circ$, there is then a unique incidence point (x, H) on the chosen branch.

Take a line L through H satisfying the following conditions: $x \notin A_L$, the intersection of L and D_a° is proper at H, and the tangent direction of L is not perpendicular to the conormal line represented by (x, H). These conditions define an open subset of the space of lines through H. By Barrett's projective Radon criterion [Bar24, Proposition 4.5], the complement of the projectivized singular support is the maximal ULA locus of the incidence coefficient. Since D_a° is disjoint from the other discriminant components and the map above is generically radicial, after replacing L by an open neighborhood V of H, the point $z = (x, H)$ is the only non-ULA point. This proves (i).

We prove (ii). Put $R_L = Rg_*G_L$. Up to the fixed Radon shift and twist, R_L is the restriction of $\mathbf{R}(i_!\mathcal{P})$ to V. By Barrett's specialization argument [Bar24, §4.7], the specialization morphism from the stalk at H to a geometric generic stalk of L is not an isomorphism. Therefore

$$R\Phi_H(R_L, \mathrm{id}_V) \neq 0.$$

Proper base change for nearby cycles in the pro-étale category gives an isomorphism

$$R\Phi_H(Rg_*G_L, \mathrm{id}_V) \xrightarrow{\sim} R\Gamma\big(g^{-1}(H), R\Phi_g(G_L)\big). \tag{6.6.1}$$

After multiplying the relevant morphisms by a power of π, take an integral model. Modulo π^m, (6.6.1) is the proper base-change isomorphism for nearby cycles. The cone of the integral comparison is derived π-complete, and all its reductions are zero. It is therefore zero by lemma 5.2. Rationalization proves (6.6.1) over E. By (i), the complex on the right is supported at z. Hence $R\Phi_z(G_L, g) \neq 0$.

Since $x \notin A_L$, the morphism p_L is an isomorphism near z. After shrinking the target around x, we may assume that i is a closed immersion. Thus $i_!\mathcal{P} = i_*\mathcal{P}$, and G_L is perverse near z. The functor $R\Phi_g[-1]$ is perverse t-exact in the constructible pro-étale category. Since its support near the fiber is $\{z\}$, it is an ordinary E-sheaf in perverse degree zero. It follows that

$$R^q\Phi_z(G_L, g) = 0 \quad (q \neq -1), \qquad R^{-1}\Phi_z(G_L, g) \neq 0,$$

which proves (ii). On the neighborhood where p_L is an isomorphism, the differential of g is the covector represented by (x, H). This proves (iii). Thus the finite-coefficient argument of [Sai17, Lemma 3.10] extends to the pro-étale category. □

Theorem 6.7 (Perverse equality)**.** *For every perverse E-sheaf $\mathcal{P}$ on a smooth k-scheme X,*

$$\mathrm{SS}_\mu(\mathcal{P}) = \mathrm{SS}(\mathcal{P}).$$

Proof. The inclusion $\mathrm{SS}_\mu(\mathcal{P}) \subset \mathrm{SS}(\mathcal{P})$ follows from Theorem 1.4. Suppose that the reverse inclusion does not hold. Since $\mathrm{SS}_\mu(\mathcal{P})$ is closed, there is an irreducible component $C_a \subset \mathrm{SS}(\mathcal{P})$ having a dense open subset disjoint from $\mathrm{SS}_\mu(\mathcal{P})$. Take a covector in this open subset and a pencil as in lemma 6.6. Then z is the only $\mathrm{SS}(G_L)$-characteristic point of g over H.

Near x, factor i as a closed immersion into an open subset of $\mathbb{P}$. The closed-immersion formula $\mathrm{SS}_\mu(i_*\mathcal{P}) = i_\circ \mathrm{SS}_\mu(\mathcal{P})$ and the local-isomorphism formula for p_L show that dg_z is not in $\mathrm{SS}_\mu(G_L)$. At every other point of $g^{-1}(V)$, (g, G_L) is ULA by lemma 6.6(i). By the test-pair criterion and the equality of weak and full ℓ-adic singular support [Bar24, Theorem 1.5(vii)], the section dg avoids $\mathrm{SS}(G_L)$ at these points. By Theorem 1.4, it also avoids $\mathrm{SS}_\mu(G_L)$. At z, the same conclusion follows from the closed-immersion formula above. Therefore the support condition (6.1.2) holds in its stronger, empty-intersection form:

$$(dg)^{-1}\,\mathrm{SS}_\mu(G_L) = \varnothing \qquad \text{on } g^{-1}(V).$$

Hence $I_{g,H}(\mathrm{CC}_\mu(G_L)) = 0$. Applying proposition 6.4 to the proper morphism g, we obtain

$$a_H(Rg_*G_L) = 0. \tag{6.7.1}$$

By lemma 6.6(i), z is the only non-ULA point over H. The vanishing-cycle triangle and (6.6.1) identify the Artin conductor with the total dimension of the vanishing cycles at z. By Saito's Milnor formula, it is also the corresponding intersection number. Thus

$$\begin{aligned} a_H(Rg_*G_L) &= \mathrm{dimtot}\, R\Phi_z(G_L, g) \\ &= \mathrm{dimtot}\, R^{-1}\Phi_z(G_L, g) \\ &= -\,(\mathrm{CC}(G_L), dg)_z > 0. \end{aligned} \tag{6.7.2}$$

The last equality is Saito's Milnor formula [Sai17, Theorem 5.9]. The sign follows from the concentration in degree -1, and the middle term is the positive total dimension of the nonzero E-representation $R^{-1}\Phi_z(G_L, g)$. This contradicts (6.7.1). Hence every irreducible component of $\mathrm{SS}(\mathcal{P})$ is contained in $\mathrm{SS}_\mu(\mathcal{P})$, and the theorem follows. For finite coefficients, if the total dimension and the residue degree are units, the same contradiction follows from corollary 6.3 and Yang–Zhao's cohomological Milnor formula. □

Remark 6.8 (Coefficient limitation)**.** Characteristic zero is used only in the last contradiction. For finite coefficients, a positive total dimension may vanish modulo ℓ. For a general derived complex, the scalar traces in different cohomological degrees may also cancel. Thus the theorem does not give the reverse inclusion for arbitrary finite-coefficient complexes.

7. Microlocal and cohomological characteristic classes

7.1. In this section, we apply Theorem 1.4 to characteristic classes. The resulting dimension estimate allows us to apply Saito's microlocal cycle formula. We first compare the microlocal image of the identity endomorphism with the construction of Kashiwara–Schapira. We then establish two zero-section formulas with support and deduce the comparison between the cohomological characteristic class and the zero-dimensional characteristic cycle.

Corollary 7.2 (Cycle-class comparison)**.** *For every constructible complex F with coefficients as in Theorem 1.4,*

$$\mathrm{CC}_\mu(F) = \mathrm{cl}(\mathrm{CC}(F)). \tag{7.2.1}$$

For a perverse E-sheaf, every irreducible component of this cycle class is supported on the common set $\mathrm{SS}_\mu(F) = \mathrm{SS}(F)$.

Proof. By Theorem 1.4, we have $\dim \mathrm{SS}_\mu(F) \leq \dim X$ for every smooth X. Hence Saito's comparison [Sai26, Proposition 2.5.2] applies and gives (7.2.1) for finite coefficients.

We now treat E-coefficients. Both CC_μ and CC are additive for distinguished triangles. For CC_μ, this is [Sai26, Proposition 2.1.7]; for CC, it follows from the Milnor characterization. By Barrett's decomposition

$$\mathrm{SS}(F) = \bigcup_i \mathrm{SS}({}^pH^iF)$$

and its refinement by the perverse Jordan–Hölder constituents [Bar24, Theorem 1.5(viii)], it is enough to prove the comparison for a perverse E-sheaf $\mathcal{P}$.

After replacing E by a finite extension, take a torsion-free perverse integral model $\mathcal{P}_0$. By [Bar24, Theorem 1.5(v)], we have

$$\mathrm{SS}(\mathcal{P}_0) = \mathrm{SS}(\mathcal{P}) \tag{7.2.2}$$

for every such model. Moreover, the perverse t-exactness of $R\Phi[-1]$ implies that $R\Phi(\mathcal{P}_0)[-1]$ is torsion-free; this is the multiplicity argument in the proof of the cited theorem. Put $\kappa_E = \mathcal{O}_E/\pi$. Let (u, f) be a generic isolated characteristic test for an irreducible component, and put

$$V_0 = R^{-1}\Phi_u(\mathcal{P}_0, f).$$

Then V_0 is a finite free $\mathcal{O}_E$-module endowed with an inertia action. Its generic and special fibers are, respectively, $R^{-1}\Phi_u(\mathcal{P}, f)$ and $R^{-1}\Phi_u(\mathcal{P}_0 \otimes^L \kappa_E, f)$. For every higher wild inertia group, the formation of invariants of V_0 commutes with both base changes. Indeed, after passing to a finite quotient, this follows from the averaging idempotent, whose denominator is a power of p and hence a unit in $\mathcal{O}_E$. Consequently the rank and every term in the Swan-conductor formula are preserved. Therefore

$$\mathrm{dimtot}_E\, R^{-1}\Phi_u(\mathcal{P}, f) = \mathrm{dimtot}_{\kappa_E}\, R^{-1}\Phi_u(\mathcal{P}_0 \otimes^L \kappa_E, f). \tag{7.2.3}$$

By the Milnor characterization, the rationalization of the integral characteristic cycle is the cycle occurring in the finite-level comparison for $\mathcal{P}_0 \otimes^L \mathcal{O}_E/\pi^m$, for every m.

For every m, apply the finite-coefficient comparison to $\mathcal{P}_0 \otimes^L \mathcal{O}_E/\pi^m$. By lemma 5.2, the resulting microlocal classes are the reductions of the integral identity-trace class $\mathrm{CC}_\mu(\mathcal{P}_0)$. Let δ_0 be the difference between this class and the $\mathcal{O}_E$-cycle class of the integral cycle above. The finite-coefficient comparison gives

$$\delta_0 \otimes^L \mathcal{O}_E/\pi^m = 0 \qquad \text{for every } m.$$

The supported cohomology complex containing δ_0 is derived π-complete. Hence $\delta_0 = 0$. After tensoring with E and applying (7.2.3), we obtain (7.2.1) for $\mathcal{P}$. By additivity, it holds for F. Notice that no arbitrary reduction is used to calculate the rational singular support.

If F is perverse, the support assertion follows from Theorem 6.7. Equivalently, the generic coefficient of each component of $\mathrm{CC}(F)$ is a positive integer by [Sai17, Proposition 4.14], and hence remains nonzero over E. □

We formulate the étale analogue of the identity-support statement of Kashiwara–Schapira [KS90, Corollary 6.1.3]. In this paragraph only, we retain the geometric generic-point factor. Put

$$\widetilde{\mathcal{M}}_F = \mathfrak{F}_{\psi,TX/X}\nu_{\Delta_X}(\mathscr{H}_F) \quad \text{on } T^*X \times_k \eta_0,$$

and let $q_\eta : T^*X \times_k \eta_0 \to T^*X$ be the projection. Thus $\mu\mathcal{H}om(F,F)$ denotes $\widetilde{\mathcal{M}}_F$ with η_0 suppressed, and

$$\mathrm{SS}_\mu(F) = \overline{q_\eta\big(\operatorname{supp}\widetilde{\mathcal{M}}_F\big)}.$$

Here and in the rest of this paragraph, projected support means its closed conical closure. This is the precise meaning of the convention that the factor η_0 is suppressed.

Proposition 7.3 (Microlocal identity section)**.** *Let F be a constructible complex on a smooth k-scheme X, with coefficients as in Theorem 1.4. The identity endomorphism of F defines a canonical morphism*

$$\widetilde{s}_F : \Lambda_{T^*X\times_k\eta_0} \longrightarrow \widetilde{\mathcal{M}}_F, \tag{7.3.1}$$

where Λ is replaced by E for an E-complex. After suppressing η_0, we denote this morphism by

$$s_F : \Lambda_{T^*X} \longrightarrow \mu\mathcal{H}om(F,F).$$

More precisely, this is shorthand for (7.3.1), whose source is $q_\eta^*\Lambda_{T^*X} = \Lambda_{T^*X\times_k\eta_0}$; no descent of $\widetilde{\mathcal{M}}_F$ along q_η is being asserted. Let $\mathcal{I}_F$ be the image of $\mathcal{H}^0(\widetilde{s}_F) : \Lambda \to \mathcal{H}^0(\widetilde{\mathcal{M}}_F)$. We define

$$\operatorname{supp}(s_F) = \overline{q_\eta(\operatorname{supp}\mathcal{I}_F)}. \tag{7.3.2}$$

Equivalently, it is the closure of the projections of the geometric points $\bar\xi$ for which the stalk class $s_{F,\bar\xi} \in H^0(\widetilde{\mathcal{M}}_{F,\bar\xi})$ is nonzero. Then

$$\operatorname{supp}(s_F) \subset \mathrm{SS}_\mu(F) \subset \mathrm{SS}(F). \tag{7.3.3}$$

Moreover, the microlocal characteristic class is the image of s_F under the microlocal evaluation morphism. If $F = \mathcal{P}$ is a perverse E-sheaf, then

$$\operatorname{supp}(s_{\mathcal{P}}) = \operatorname{supp}\mu\mathcal{H}om(\mathcal{P},\mathcal{P}) = \mathrm{SS}_\mu(\mathcal{P}) = \mathrm{SS}(\mathcal{P}). \tag{7.3.4}$$

Proof. We first work with finite coefficients. Put

$$\mathscr{H}_F = R\mathcal{H}om(\mathrm{pr}_1^*F, \mathrm{pr}_2^!F), \qquad K_X = K_{X/k}.$$

Since F has finite Tor-dimension, it is dualizable. Smooth purity for the second projection gives

$$\mathscr{H}_F \simeq \mathbb{D}_X F \boxtimes F, \qquad \Delta_X^*\mathscr{H}_F \simeq \mathbb{D}_X F \otimes^L F. \tag{7.3.5}$$

Under the canonical isomorphism

$$R\mathcal{H}om(F,F) \xrightarrow{\sim} \Delta_X^!\mathscr{H}_F, \tag{7.3.6}$$

the identity of F defines $1_F : \Lambda_X \to \Delta_X^!\mathscr{H}_F$. Under the second isomorphism in (7.3.5), the evaluation $\mathbb{D}_X F \otimes^L F \to K_X$ defines $\mathrm{ev}_F : \Delta_X^*\mathscr{H}_F \to K_X$. Together with the canonical natural transformation $\Delta_X^! \to \Delta_X^*$, these morphisms give

$$\Lambda_X \xrightarrow{1_F} \Delta_X^!\mathscr{H}_F \longrightarrow \Delta_X^*\mathscr{H}_F \xrightarrow{\mathrm{ev}_F} K_X. \tag{7.3.7}$$

Taking the adjoints of the first and last morphisms in (7.3.7), we obtain

$$\Delta_{X!}\Lambda_X \xrightarrow{\widetilde{1}_F} \mathscr{H}_F \xrightarrow{\widetilde{\mathrm{ev}}_F} \Delta_{X*}K_X. \tag{7.3.8}$$

The composite obtained from (7.3.8) by applying $\Delta_X^!$ on the left and Δ_X^* on the right is (7.3.7). This observation fixes the two adjunctions and the direction of all the following morphisms.

Let $0_X : X \hookrightarrow TX$ be the zero section. Proper base change on the deformation to the normal bundle gives canonical isomorphisms

$$\nu_{\Delta_X}(\Delta_{X!}\Lambda_X) \xrightarrow{\sim} 0_{X!}\Lambda_X, \qquad \nu_{\Delta_X}(\Delta_{X*}K_X) \xrightarrow{\sim} 0_{X*}K_X. \tag{7.3.9}$$

Consequently, specialization of (7.3.8) is the sequence

$$0_{X!}\Lambda_X \longrightarrow \nu_{\Delta_X}(\mathscr{H}_F) \longrightarrow 0_{X*}K_X \tag{7.3.10}$$

on $TX \times_k \eta_0$. Let $e_X^\vee : T^*X \to X$ be the projection. The zero-section formulas for the unshifted Fourier–Deligne transform are

$$\mathfrak{F}_{\psi,TX/X}(0_{X!}\Lambda_X) \xrightarrow{\sim} e_X^{\vee *}\Lambda_X, \qquad \mathfrak{F}_{\psi,TX/X}(0_{X*}K_X) \xrightarrow{\sim} e_X^{\vee *}K_X. \tag{7.3.11}$$

Here $0_{X!} = 0_{X*}$ because 0_X is a closed immersion. Applying Fourier transformation to (7.3.10) and using (7.3.11), we obtain the commutative diagram

$$\begin{tikzcd} \mathfrak{F}_\psi(0_{X!}\Lambda_X) \arrow[r] \arrow[d, "\sim"'] & \mathfrak{F}_\psi\nu_{\Delta_X}(\mathscr{H}_F) \arrow[r] \arrow[d, equal] & \mathfrak{F}_\psi(0_{X*}K_X) \arrow[d, "\sim"] \\ \Lambda_{T^*X} \arrow[r, "s_F"'] & \mu\mathcal{H}om(F,F) \arrow[r, "\mathrm{ev}_{\mu,F}"'] & e_X^{\vee *}K_X. \end{tikzcd} \tag{7.3.12}$$

The factors η_0 in (7.3.12) are suppressed. The bottom row is the pair of structural morphisms in [Sai26, (2.10)]; its supported composite is Saito's microlocal characteristic class [Sai26, Definition 2.1.4].

We make the supported composite explicit. Put $C_F = \mathrm{SS}_\mu(F)$ and let $i_F : C_F \hookrightarrow T^*X$ be the closed immersion. Write $\mathcal{M}_F = \mu\mathcal{H}om(F,F)$ and $B_X = e_X^{\vee *}K_X$. Since $\mathcal{M}_F$ is supported on C_F, the adjunction morphism is an isomorphism

$$Ri_{F*}i_F^*\mathcal{M}_F \xrightarrow{\sim} \mathcal{M}_F. \tag{7.3.13}$$

The first morphism in the bottom row of (7.3.12) restricts to $i_F^*s_F : \Lambda_{C_F} \to i_F^*\mathcal{M}_F$. By (7.3.13) and the adjunction $(Ri_{F*}, i_F^!)$, the evaluation morphism has a unique adjoint $\widetilde{\mathrm{ev}}_{\mu,F} : i_F^*\mathcal{M}_F \to i_F^!B_X$. Thus we have a canonical composite

$$\Lambda_{C_F} \xrightarrow{i_F^*s_F} i_F^*\mathcal{M}_F \xrightarrow{\widetilde{\mathrm{ev}}_{\mu,F}} i_F^!B_X. \tag{7.3.14}$$

Using

$$\mathrm{Hom}_{D(C_F)}(\Lambda_{C_F}, i_F^!B_X) \xrightarrow{\sim} H^0_{C_F}(T^*X, B_X), \tag{7.3.15}$$

the class of (7.3.14) is

$$\mathrm{CC}_\mu(F) \in H^0_{C_F}(T^*X, e_X^{\vee *}K_X). \tag{7.3.16}$$

In particular, on every open subset $V \subset T^*X$ on which s_F is zero, the restriction of $\mathrm{CC}_\mu(F)$ is zero. In this precise sense,

$$\mathrm{supp}(\mathrm{CC}_\mu(F)) \subset \mathrm{supp}(s_F) \subset \mathrm{supp}\,\mu\mathcal{H}om(F,F) = \mathrm{SS}_\mu(F). \tag{7.3.17}$$

The last inclusion in (7.3.3) is Theorem 1.4.

For E-coefficients, all objects and morphisms above are defined directly in the constructible pro-étale category. To verify their compatibility, choose an integral model F_0 and put $F_m = F_0 \otimes^L_{\mathcal{O}_E} \mathcal{O}_E/\pi^m$. For every m, the reduction of (7.3.12) is the diagram constructed above for F_m.

These diagrams commute with the transition morphisms. Derived completeness therefore gives the corresponding diagram over $\mathcal{O}_E$, independently of the chosen model after rationalization. Inverting π gives (7.3.12)– (7.3.16) over E.

Let now $F = \mathcal{P}$ be perverse with E-coefficients. By Theorem 6.7,

$$C_{\mathcal{P}} = \mathrm{SS}_\mu(\mathcal{P}) = \mathrm{SS}(\mathcal{P}).$$

Write

$$C_{\mathcal{P}} = \bigcup_a C_a, \qquad \mathrm{CC}(\mathcal{P}) = \sum_a m_a[C_a]. \tag{7.3.18}$$

The multiplicities m_a are positive integers by [Sai17, Proposition 4.14]. By corollary 7.2,

$$\mathrm{CC}_\mu(\mathcal{P}) = \sum_a m_a \,\mathrm{cl}[C_a]. \tag{7.3.19}$$

Fix a component C_a with generic point ξ_a. Since T^*X is smooth and C_a has dimension $\dim X$, we may choose an open neighborhood V_a of ξ_a such that

$$V_a \cap C_{\mathcal{P}} = V_a \cap C_a$$

and $V_a \cap C_a$ is regular of codimension $\dim X$ in V_a. Choose a geometric point $\bar{\xi}_a$ above ξ_a, let V_a^{sh} be the strict localization of V_a at $\bar{\xi}_a$, and put $C_a^{\mathrm{sh}} = (C_a \cap V_a) \times_{V_a} V_a^{\mathrm{sh}}$. Absolute purity identifies the local supported group at $\bar{\xi}_a$ with the one-dimensional E-space generated by the cycle class:

$$H^0_{C_a^{\mathrm{sh}}}(V_a^{\mathrm{sh}}, e_X^{\vee *}K_X) \xrightarrow{\sim} E \cdot \mathrm{cl}[C_a^{\mathrm{sh}}]. \tag{7.3.20}$$

The restriction of (7.3.19) to this group is $m_a\,\mathrm{cl}[C_a^{\mathrm{sh}}]$. It is nonzero because $m_a > 0$ and E has characteristic zero.

Suppose that ξ_a were not contained in $\mathrm{supp}(s_{\mathcal{P}})$. Constructibility of the image sheaf in (7.3.2) would give, after shrinking V_a, a neighborhood on which $s_{\mathcal{P}}$ is zero. By (7.3.14), the restriction of $\mathrm{CC}_\mu(\mathcal{P})$ to the left hand side of (7.3.20) would then be zero, a contradiction. Therefore $\xi_a \in \mathrm{supp}(s_{\mathcal{P}})$. Since the latter is closed and the argument applies to every a, we obtain

$$C_{\mathcal{P}} \subset \mathrm{supp}(s_{\mathcal{P}}).$$

Together with (7.3.3) and the definition of $C_{\mathcal{P}}$, this proves (7.3.4). □

Remark 7.4 (Comparison with Kashiwara–Schapira)**.** In the transcendental setting, the identity section is obtained from the isomorphism

$$R\,\mathrm{Hom}(F,F) \simeq R\pi_*\mu hom(F,F)$$

and its support is the singular support [KS90, Corollary 6.1.3]. In the étale setting, $\widetilde{\mathcal{M}}_F$ is defined over the geometric generic point of the henselized deformation parameter and carries its monodromy action. A literal push-forward to X would therefore require an additional choice of how to treat this monodromy. We avoid such a choice. The diagonal construction (7.3.7), followed by the canonical specialization and Fourier diagram (7.3.12), transports the identity endomorphism directly. The supported class (7.3.16) is the precise étale analogue of [KS90, Definition 9.4.1]. It is a class with coefficients in Λ or E; for finite Λ, it records the integral characteristic-cycle multiplicities only after reduction to Λ.

Remark 7.5 (Finite-coefficient trace loss)**.** For arbitrary coefficients, Proposition 7.3 gives only the inclusions (7.3.3). The trace class need not detect the support of the identity section. For example, let $\Lambda = \mathbf{F}_\ell$, let X be connected and smooth of dimension n, and put $F = \Lambda_X^{\oplus \ell}$. Then

$$\mathrm{SS}_\mu(F) = \mathrm{supp}(s_F) = T^*_X X, \qquad \mathrm{CC}_\mu(F) = (-1)^n \ell\,[T^*_X X] = 0$$

with $\mathbf{F}_\ell$-coefficients. Indeed, the locally constant calculation identifies the restriction of s_F to the zero section with id_F, whereas the evaluation sends it to the scalar $\mathrm{Tr}(\mathrm{id}_F) = \ell = 0$; see [Sai26, Lemmas 2.1.2 and 2.1.5]. Thus the vanishing of $\mathrm{CC}_\mu(F)$ is not the vanishing of s_F.

There is also a characteristic-zero cancellation for a nonperverse complex. If L is a nonzero locally constant E-sheaf and $F = L \oplus L[1]$, additivity gives

$$\mathrm{CC}_\mu(F) = \mathrm{CC}_\mu(L) + \mathrm{CC}_\mu(L[1]) = 0,$$

while the restriction of s_F to the zero section is $(\mathrm{id}_L, \mathrm{id}_{L[1]})$ and is nonzero. Neither example is a counterexample to $\mathrm{supp}(s_F) = \mathrm{SS}_\mu(F)$. They show that the trace class cannot prove this equality without a positivity hypothesis. We do not assert the reverse inclusions in (7.3.3) for arbitrary finite coefficients or for a general derived E-complex.

7.6. Let $i : Z \hookrightarrow M$ be a closed immersion with M smooth. For a finite-type k-scheme T, put $K_{T/k} = Ra_T^!\Lambda$. By Grothendieck duality, we have $K_{Z/k} \simeq i^! K_{M/k}$. Excision gives an isomorphism

$$\iota_Z : H^0(Z, K_{Z/k}) \xrightarrow{\sim} H^0_Z(M, K_{M/k}). \tag{7.6.1}$$

Let $0_M : M \hookrightarrow T^*M$ be the zero section. For $F \in D^b_{\mathrm{ctf}}(Z, \Lambda)$ and $G = i_*F$, both $\mathrm{SS}_\mu(G)$ and $\mathrm{SS}(G)$ lie over Z. Hence the zero-section pullback refines to a morphism

$$0^*_{M,Z} : H^0_{\mathrm{SS}(G)}(T^*M, e_M^{\vee *} K_{M/k}) \longrightarrow H^0_Z(M, K_{M/k}). \tag{7.6.2}$$

Lemma 7.7 (Microlocal zero-section formula)**.** *Under* (7.6.1),

$$0^*_{M,Z}\,\mathrm{CC}_\mu(i_*F) = \iota_Z(C_{Z/k}(F)). \tag{7.7.1}$$

Proof. Put $j : M \setminus Z \hookrightarrow M$. By the localization triangle and proper duality, we have canonical isomorphisms

$$R\Gamma_Z K_{M/k} = \mathrm{fib}(K_{M/k} \to Rj_*j^*K_{M/k}) \xrightarrow{\sim} i_*i^!K_{M/k} = i_*K_{Z/k}, \qquad \mathbb{D}_M(i_*F) \xrightarrow{\sim} i_*\mathbb{D}_Z F. \tag{7.7.2}$$

Therefore the evaluation morphism for $G = i_*F$ has a canonical factorization with support:

$$\mathbb{D}_M G \otimes^L G \longrightarrow i_*(\mathbb{D}_Z F \otimes^L F) \xrightarrow{i_*\,\mathrm{ev}_F} i_*K_{Z/k} \xrightarrow{\sim} R\Gamma_Z K_{M/k} \longrightarrow K_{M/k}. \tag{7.7.3}$$

The first morphism is induced by the projection formula. After applying i^*, it becomes the canonical tensor-product morphism on Z and hence is an isomorphism.

We next refine Saito's zero-section diagram with support. Put

$$\mathscr{H}_G = R\mathcal{H}om(\mathrm{pr}_1^* G, \mathrm{pr}_2^! G), \qquad \mathscr{H}_F = R\mathcal{H}om(\mathrm{pr}_1^* F, \mathrm{pr}_2^! F).$$

The cartesian diagram of the diagonals and proper base change give the canonical isomorphisms

$$\mathscr{H}_G \simeq R(i \times i)_*\mathscr{H}_F, \qquad \Delta_M^!\mathscr{H}_G \simeq i_*\Delta_Z^!\mathscr{H}_F, \qquad \Delta_M^*\mathscr{H}_G \simeq i_*\Delta_Z^*\mathscr{H}_F.$$

Under these isomorphisms, the full faithfulness of i_* identifies id_G with $i_*(\mathrm{id}_F)$, and the projection formula identifies the evaluation morphisms. We therefore obtain a commutative diagram in $D(M, \Lambda)$:

$$\begin{tikzcd}
\Lambda_M \arrow[r, "1_G"] \arrow[d] & \Delta_M^!\mathscr{H}_G \arrow[r] \arrow[d, "\sim"'] & \Delta_M^*\mathscr{H}_G \arrow[r, "\mathrm{ev}_{G,Z}"] \arrow[d, "\sim"'] & R\Gamma_Z K_{M/k} \arrow[d, "\sim"] \\
i_*\Lambda_Z \arrow[r, "i_*1_F"'] & i_*\Delta_Z^!\mathscr{H}_F \arrow[r] & i_*\Delta_Z^*\mathscr{H}_F \arrow[r, "i_*\,\mathrm{ev}_F"'] & i_*K_{Z/k}.
\end{tikzcd} \tag{7.7.4}$$

Here $\mathrm{ev}_{G,Z}$ is the supported evaluation morphism in (7.7.3). Its composite with $R\Gamma_Z K_{M/k} \to K_{M/k}$ is the evaluation morphism of G.

We compare (7.7.4) with microlocalization. Recall the three morphisms in the proof of [Sai26, Lemma 2.1.6]. First, Fourier base change gives $0_M^*\mathfrak{F}_\psi \simeq e_!$ for $e : TM \to M$. Second, the unit of specialization identifies pullback by the zero section of TM with restriction to the diagonal. Third, the specialization of evaluation is the evaluation after this restriction. These morphisms are natural in $\mathscr{H}_G$. Replacing the last evaluation morphism by $\mathrm{ev}_{G,Z}$, we obtain a commutative diagram with support:

$$\begin{tikzcd} \Lambda_M \arrow[r] \arrow[d, equal] & 0_M^* \mu\mathcal{H}om(G,G) \arrow[r, "0_M^* \mathrm{ev}_{\mu,Z}"] \arrow[d, "\sigma_G"] & R\Gamma_Z K_{M/k} \arrow[d] \\ \Lambda_M \arrow[r, "1_G"'] & \Delta_M^! \mathscr{H}_G \arrow[r] & K_{M/k}, \end{tikzcd} \tag{7.7.5}$$

The lower composite is the upper row of (7.7.4) followed by the forget-support morphism. The morphism σ_G is the composite of Fourier base change and the zero-section unit for specialization. The left square is the Fourier–specialization unit square, and the right square follows from the naturality of evaluation. Thus (7.7.5) is Saito's zero-section diagram before forgetting support.

Combining (7.7.4) and (7.7.5), we obtain the factorization

$$\Lambda_M \longrightarrow i_*\Lambda_Z \xrightarrow{i_*C_{Z/k}(F)} i_*K_{Z/k} \simeq R\Gamma_Z K_{M/k} \longrightarrow K_{M/k}. \tag{7.7.6}$$

By excision, the part of (7.7.6) ending in $R\Gamma_Z K_{M/k}$ represents $\iota_Z(C_{Z/k}(F))$. On the other hand, (7.7.5) identifies it with $0_{M,Z}^* \mathrm{CC}_\mu(G)$. This proves (7.7.1). Notice that the argument is made in $R\Gamma_Z K_{M/k}$. It does not use the forget-support morphism

$$H_Z^0(M, K_{M/k}) \longrightarrow H^0(M, K_{M/k}),$$

which is not injective in general. □

Lemma 7.8 (Cycle-theoretic zero-section formula)**.** *Let* cl_M *be the cycle-class map with support on* T^*M*. Then*

$$0_{M,Z}^* \mathrm{cl}_M(\mathrm{CC}(i_*F)) = \iota_Z\big(\mathrm{cl}_Z(cc_{Z,0}(F))\big). \tag{7.8.1}$$

Proof. Put $A = \mathrm{CC}(i_*F)$. It is a cycle on T^*M whose base is contained in Z. Since the zero section is a regular immersion, the compatibility of the cycle-class map with refined Gysin pullback gives a commutative diagram

$$\begin{tikzcd} CH_{\dim M}(\mathrm{SS}(i_*F)) \arrow[r, "\mathrm{cl}_M"] \arrow[d, "0_M^!"'] & H^0_{\mathrm{SS}(i_*F)}(T^*M, e_M^{\vee *} K_{M/k}) \arrow[d, "0_{M,Z}^*"] \\ CH_0(Z) \arrow[r, "\iota_Z \circ \mathrm{cl}_Z"'] & H_Z^0(M, K_{M/k}). \end{tikzcd}$$

The lower horizontal morphism is the cycle-class map followed by the excision isomorphism, since the intersection is supported on Z. Applying the diagram to A, we obtain

$$\begin{aligned} 0_{M,Z}^* \mathrm{cl}_M(\mathrm{CC}(i_*F)) &= \iota_Z \, \mathrm{cl}_Z(0_M^! \, \mathrm{CC}(i_*F)) \\ &= \iota_Z \, \mathrm{cl}_Z(cc_{Z,0}(F)). \end{aligned}$$

The second equality follows from Saito's zero-section description of the zero-dimensional part $cc_{Z,0}(F)$ of the total characteristic class [Sai17, Definition 6.7.2 and Lemma 6.9.1]. □

Theorem 7.9 (Saito's conjecture, [Sai17, Conjecture 6.8])**.** *Let Z be a finite-type k-scheme admitting a closed immersion into a smooth finite-type k-scheme. For every* $F \in D^b_{\mathrm{ctf}}(Z, \Lambda)$,

$$C_{Z/k}(F) = \mathrm{cl}_Z(cc_{Z,0}(F)) \quad \text{in } H^0(Z, K_{Z/k}). \tag{7.9.1}$$

Proof. Take a closed immersion $i : Z \hookrightarrow M$ with M smooth, and put $G = i_*F$. By Theorem 1.4, we have

$$\mathrm{SS}_\mu(G) \subset \mathrm{SS}(G).$$

By Beilinson's dimension theorem, $\dim \mathrm{SS}(G) \le \dim M$. Hence $\dim \mathrm{SS}_\mu(G) \le \dim M$. Applying [Sai26, Proposition 2.5.2], we obtain

$$\mathrm{CC}_\mu(G) = \mathrm{cl}_M(\mathrm{CC}(G)) \quad \text{in } H^0_{\mathrm{SS}(G)}(T^*M, e_M^{\vee *} K_{M/k}). \tag{7.9.2}$$

Apply $0^*_{M,Z}$ to (7.9.2). By lemmas 7.7 and 7.8, the two sides are, respectively,

$$\iota_Z(C_{Z/k}(F)) \quad \text{and} \quad \iota_Z(\mathrm{cl}_Z(cc_{Z,0}(F))).$$

Since ι_Z is an isomorphism, we obtain (7.9.1).

Suppose that Λ does not contain the values of ψ. Take a finite faithfully flat local extension $\Lambda \to \Lambda'$ containing them. Microlocalization, the two characteristic classes, the cycle-class map, and zero-section pullback commute with this extension. The equality over Λ' therefore descends to Λ by faithful flatness. □

Remark 7.10 (Logical scope)**.** The reverse inclusion for perverse sheaves and the characteristic-class theorem are two independent consequences of Theorem 1.4. The first uses the microlocal conductor identity and characteristic-zero positivity to detect every component of the singular support. Under the numerical condition of corollary 6.3, the finite-coefficient NA class gives a parallel obstruction. The characteristic-class theorem uses only the forward inclusion, which supplies the dimension hypothesis in Saito's microlocal cycle formula, and the two zero-section formulas with support. It therefore applies to every finite-Tor-dimension constructible complex with finite local coefficients, including the case where Z is singular.

8. External products

8.1. Let X and Y be smooth k-schemes, and let

$$F \in D^b_{\mathrm{ctf}}(X, \Lambda), \qquad G \in D^b_{\mathrm{ctf}}(Y, \Lambda).$$

We use the same notation for constructible complexes with coefficients in a finite extension $E/\mathbf{Q}_\ell$ on the pro-étale site. We identify

$$T^*(X \times Y) = T^*X \times T^*Y \tag{8.1.1}$$

by the canonical decomposition $\Omega^1_{X\times Y/k} \simeq \mathrm{pr}_X^* \Omega^1_{X/k} \oplus \mathrm{pr}_Y^* \Omega^1_{Y/k}$.

In the proofs below, we retain the geometric generic-point factor. For specialized complexes A on $TX \times \eta_0$ and B on $TY \times \eta_0$, put

$$A \boxtimes_{\eta_0} B = \mathrm{pr}_{TX}^* A \otimes^L \mathrm{pr}_{TY}^* B$$

on $TX \times TY \times \eta_0$. In the statements, we suppress this factor according to the convention in (1.1.2).

For cohomology classes with support, $\alpha \boxtimes \beta$ denotes the cohomological external product followed by the canonical isomorphism

$$K_{X/k} \boxtimes K_{Y/k} \xrightarrow{\sim} K_{X\times Y/k}. \tag{8.1.2}$$

Lemma 8.2 (Products of specialization kernels)**.** *Let*

$$\mathscr{H}_F = R\mathcal{H}om(p_{1,X}^* F, p_{2,X}^! F), \qquad \mathscr{H}_G = R\mathcal{H}om(p_{1,Y}^* G, p_{2,Y}^! G).$$

Under the permutation isomorphism

$$\sigma : (X \times Y)^2 \xrightarrow{\sim} X^2 \times Y^2, \qquad ((x_1, y_1), (x_2, y_2)) \longmapsto ((x_1, x_2), (y_1, y_2)),$$

there are canonical isomorphisms

$$\mathscr{H}_{F\boxtimes G} \xrightarrow{\sim} \sigma^*(\mathscr{H}_F \boxtimes \mathscr{H}_G), \tag{8.2.1}$$

$$\nu_{\Delta_{X\times Y}}(\mathscr{H}_{F\boxtimes G}) \xrightarrow{\sim} \nu_{\Delta_X}(\mathscr{H}_F) \boxtimes_{\eta_0} \nu_{\Delta_Y}(\mathscr{H}_G). \tag{8.2.2}$$

In (8.2.2)*, the normal bundle on the left is identified with $TX \times TY$.*

Proof. The isomorphism σ identifies $\Delta_{X\times Y}$ with $\Delta_X \times \Delta_Y$. Put

$$A_X = p_{1,X}^*F, \quad B_X = p_{2,X}^!F, \qquad A_Y = p_{1,Y}^*G, \quad B_Y = p_{2,Y}^!G.$$

By the Künneth isomorphisms for ordinary and exceptional pullback, we have

$$\mathrm{pr}_1^*(F \boxtimes G) \simeq \sigma^*(A_X \boxtimes A_Y), \qquad \mathrm{pr}_2^!(F \boxtimes G) \simeq \sigma^*(B_X \boxtimes B_Y). \tag{8.2.3}$$

The second isomorphism in (8.2.3) is induced by

$$(p_{2,X} \times p_{2,Y})^!(- \boxtimes -) \simeq p_{2,X}^!(-) \boxtimes p_{2,Y}^!(-).$$

Since F and G are constructible of finite Tor-dimension, the complexes A_X and A_Y are dualizable for the derived tensor product. Put $A_X^\vee = R\mathcal{H}om(A_X, \Lambda_{X^2})$, and define $A_Y^\vee$ similarly. Then

$$\begin{aligned} R\mathcal{H}om(A_X, B_X) \boxtimes R\mathcal{H}om(A_Y, B_Y) &\simeq (A_X^\vee \otimes^L B_X) \boxtimes (A_Y^\vee \otimes^L B_Y) \\ &\simeq (A_X \boxtimes A_Y)^\vee \otimes^L (B_X \boxtimes B_Y) \\ &\simeq R\mathcal{H}om(A_X \boxtimes A_Y, B_X \boxtimes B_Y). \end{aligned} \tag{8.2.4}$$

The morphisms in this chain are the canonical symmetry and associativity constraints. Pulling back by σ and applying (8.2.3), we obtain (8.2.1). The construction is functorial in F and G.

We next compare the deformation spaces. Let I_X and I_Y denote the ideals of Δ_X and Δ_Y. The ideal of $\Delta_X \times \Delta_Y$ in $X^2 \times Y^2$ is

$$I = I_X \boxtimes \mathcal{O}_{Y^2} + \mathcal{O}_{X^2} \boxtimes I_Y.$$

The extended Rees algebra of I satisfies

$$\begin{aligned} \mathcal{O}_{X^2\times Y^2}[\tau, I\tau^{-1}] &= \mathcal{O}_{X^2\times Y^2}[\tau, I_X\tau^{-1}, I_Y\tau^{-1}] \\ &\simeq \mathcal{O}_{X^2}[\tau, I_X\tau^{-1}] \otimes_{k[\tau]} \mathcal{O}_{Y^2}[\tau, I_Y\tau^{-1}]. \end{aligned} \tag{8.2.5}$$

On an affine product $\operatorname{Spec} A \times \operatorname{Spec} B$, write $I_X \subset A$ and $I_Y \subset B$ for the two diagonal ideals. Then

$$\big(I_X \otimes B + A \otimes I_Y\big)^n = \sum_{a+b=n} I_X^a \otimes I_Y^b. \tag{8.2.6}$$

Consequently, both sides of (8.2.5) are the subalgebra of $(A \otimes_k B)[\tau, \tau^{-1}]$ generated by $(A \otimes_k B)[\tau]$, $I_X\tau^{-1} \otimes B$, and $A \otimes I_Y\tau^{-1}$. This proves the isomorphism on affine products. Since the construction is compatible with restriction, the local isomorphisms glue. By (2.1.1), we obtain a canonical isomorphism over $\mathbf{A}^1$

$$\mathscr{D}_{\Delta_{X\times Y}}((X\times Y)^2) \xrightarrow{\sim} \mathscr{D}_{\Delta_X}(X^2) \times_{\mathbf{A}^1} \mathscr{D}_{\Delta_Y}(Y^2). \tag{8.2.7}$$

Its restriction to the special fiber is the canonical isomorphism

$$N_{\Delta_{X\times Y}/(X\times Y)^2} \xrightarrow{\sim} N_{\Delta_X/X^2} \times N_{\Delta_Y/Y^2} \simeq TX \times TY. \tag{8.2.8}$$

Let j_X, j_Y, j_{XY} be the open immersions of the punctured fibers of the three deformation spaces, and let s_X, s_Y, s_{XY} be the closed immersions of their special fibers. The isomorphisms on the generic and special fibers give a commutative diagram

$$\begin{tikzcd} (X\times Y)^2 \times \mathbf{G}_m \arrow[r, "j_{XY}"] \arrow[d, "\sigma\times 1"'] & \mathscr{D}_{\Delta_{X\times Y}}((X\times Y)^2) \arrow[d, "\sim"] & TX \times TY \arrow[l, "s_{XY}"'] \arrow[d, "\sim"] \\ (X^2\times \mathbf{G}_m) \times_{\mathbf{G}_m} (Y^2 \times \mathbf{G}_m) \arrow[r, "j_X\times j_Y"'] & \mathscr{D}_{\Delta_X}(X^2) \times_{\mathbf{A}^1} \mathscr{D}_{\Delta_Y}(Y^2) & (TX)\times(TY) \arrow[l, "s_X\times s_Y"] \end{tikzcd} \tag{8.2.9}$$

After base change to the henselization of $\mathbf{A}^1$ at 0, (8.2.9) induces a nearby-cycle Künneth morphism

$$\kappa_{F,G} : \nu_{\Delta_X}(\mathscr{H}_F) \boxtimes_{\eta_0} \nu_{\Delta_Y}(\mathscr{H}_G) \longrightarrow \nu_{\Delta_X \times \Delta_Y}(\mathscr{H}_F \boxtimes \mathscr{H}_G). \tag{8.2.10}$$

Under (2.1.2), this morphism is induced by the canonical morphism from the external product of $s_X^* Rj_{X*}$ and $s_Y^* Rj_{Y*}$ to $(s_X \times s_Y)^* R(j_X \times j_Y)_*$. The nearby-cycle Künneth theorem [LZ22, Proposition 3.1] asserts that $\kappa_{F,G}$ is an isomorphism. Define (8.2.2) as its inverse, transported by (8.2.1), (8.2.7), and (8.2.8). This determines the direction and the canonicity of the isomorphism. Since the two factors are pulled back from the same geometric generic point, the inertia action on their external product is the diagonal action.

For E-coefficients, we work in the constructible pro-étale category. After replacing E by a finite extension, take integral models F_0, G_0 and put $F_m = F_0 \otimes^L_{\mathcal{O}_E} \mathcal{O}_E/\pi^m$ and $G_m = G_0 \otimes^L_{\mathcal{O}_E} \mathcal{O}_E/\pi^m$. The kernel isomorphisms and the morphisms κ_{F_m,G_m} are compatible with the transition maps. Passing to the derived inverse limit gives the isomorphisms for F_0, G_0. After inverting π, we obtain (8.2.1) and (8.2.2) over E. Since every morphism in the construction is canonical, the result descends from the finite extension. □

Lemma 8.3 (Fourier Künneth formula)**.** *Let $V \to S$ and $W \to T$ be vector bundles. For constructible complexes A on V and B on W, there is a canonical isomorphism*

$$\mathfrak{F}_{\psi, V\times W/S\times T}(A \boxtimes B) \xrightarrow{\sim} \mathfrak{F}_{\psi,V/S}(A) \boxtimes \mathfrak{F}_{\psi,W/T}(B). \tag{8.3.1}$$

The same assertion holds for external products over η_0.

Proof. Let $q_{1,V}, q_{2,V}$ be the projections from $V \times_S V^\vee$, and define $q_{1,W}, q_{2,W}$ similarly. Reordering the factors gives a canonical isomorphism

$$(V \times W) \times_{S\times T} (V^\vee \times W^\vee) \xrightarrow{\sim} (V \times_S V^\vee) \times (W \times_T W^\vee). \tag{8.3.2}$$

Under (8.3.2), the projections for the product bundle are identified with $q_{1,V} \times q_{1,W}$ and $q_{2,V} \times q_{2,W}$. Hence

$$(q_{1,V} \times q_{1,W})^*(A \boxtimes B) \simeq q_{1,V}^* A \boxtimes q_{1,W}^* B. \tag{8.3.3}$$

The pairing for the product bundle is

$$\langle (v,w), (\xi,\eta)\rangle = \langle v, \xi\rangle + \langle w, \eta\rangle. \tag{8.3.4}$$

Let $m : \mathbf{A}^1 \times \mathbf{A}^1 \to \mathbf{A}^1$ be the addition morphism. The identity $\psi(a+b) = \psi(a)\psi(b)$ defines a multiplicative isomorphism $m^*\mathcal{L}_\psi \simeq \mathcal{L}_\psi \boxtimes \mathcal{L}_\psi$. Pulling it back by the pairings, we obtain a canonical isomorphism

$$\mathcal{L}_\psi(\langle (v,w),(\xi,\eta)\rangle) \xrightarrow{\sim} \mathcal{L}_\psi(\langle v,\xi\rangle) \boxtimes \mathcal{L}_\psi(\langle w,\eta\rangle). \tag{8.3.5}$$

Put

$$K_V = q_{1,V}^* A \otimes \mathcal{L}_\psi(\langle v,\xi\rangle), \qquad K_W = q_{1,W}^* B \otimes \mathcal{L}_\psi(\langle w,\eta\rangle).$$

Combining (8.3.3) and (8.3.5) with the symmetry constraint for the derived tensor product, we obtain

$$q_1^*(A \boxtimes B) \otimes \mathcal{L}_\psi(\langle (v,w),(\xi,\eta)\rangle) \xrightarrow{\sim} K_V \boxtimes K_W. \tag{8.3.6}$$

The Künneth isomorphism for direct image with proper support is

$$R(q_{2,V} \times q_{2,W})_!(K_V \boxtimes K_W) \xrightarrow{\sim} Rq_{2,V!}K_V \boxtimes Rq_{2,W!}K_W. \tag{8.3.7}$$

Applying (8.3.7) to (8.3.6), we obtain (8.3.1). The construction is natural in A and B.

For complexes on $V \times \eta_0$ and $W \times \eta_0$, replace the ordinary external products in the preceding proof by fiberwise external products over η_0. The Artin–Schreier isomorphism and the $R(-)_!$ Künneth morphism commute with this base change. This proves the last assertion. □

Theorem 8.4 (Microlocal Künneth formula)**.** *After the identification* (8.1.1)*, there is a canonical isomorphism*

$$\mu\mathcal{H}om(F\boxtimes G, F\boxtimes G)\xrightarrow{\sim}\mu\mathcal{H}om(F,F)\boxtimes\mu\mathcal{H}om(G,G). \tag{8.4.1}$$

Consequently,

$$\mathrm{SS}_\mu(F\boxtimes G)=\mathrm{SS}_\mu(F)\times\mathrm{SS}_\mu(G), \tag{8.4.2}$$

and the identity/evaluation class satisfies

$$\mathrm{CC}_\mu(F\boxtimes G)=\mathrm{CC}_\mu(F)\boxtimes\mathrm{CC}_\mu(G). \tag{8.4.3}$$

These assertions also hold for constructible E-complexes on the pro-étale site.

Proof. By the definition of $\mu\mathcal{H}om$ and lemmas 8.2 and 8.3, we have

$$\begin{aligned}\mu\mathcal{H}om(F\boxtimes G, F\boxtimes G)&=\mathfrak{F}_{\psi,T(X\times Y)/(X\times Y)}\nu_{\Delta_{X\times Y}}(\mathscr{H}_{F\boxtimes G})\\&\simeq\mathfrak{F}_{\psi,TX\times TY/(X\times Y)}\big(\nu_{\Delta_X}(\mathscr{H}_F)\boxtimes_{\eta_0}\nu_{\Delta_Y}(\mathscr{H}_G)\big)\\&\simeq\mathfrak{F}_{\psi,TX/X}\nu_{\Delta_X}(\mathscr{H}_F)\boxtimes_{\eta_0}\mathfrak{F}_{\psi,TY/Y}\nu_{\Delta_Y}(\mathscr{H}_G)\\&=\mu\mathcal{H}om(F,F)\boxtimes_{\eta_0}\mu\mathcal{H}om(G,G).\end{aligned}$$

The first isomorphism is the composite of (8.2.1), the product identification of deformation spaces (8.2.7), and the inverse of the nearby-cycle Künneth map (8.2.10). The second isomorphism is (8.3.1). Each morphism is functorial in F and G. After suppressing η_0, we obtain (8.4.1).

There is no shift in (8.4.1), since we use the unshifted transformation (2.1.3). For the normalized transformations, the shifts on the right add to the shift by $\operatorname{rank}TX+\operatorname{rank}TY$ on the left. Moreover, the permutations of tensor factors use the symmetry constraint of the derived category. Thus the Koszul signs are contained in the canonical isomorphism, and there is no additional sign.

We prove (8.4.2). Let A and B be constructible complexes of finite Tor-dimension on two k-schemes. We claim that

$$\operatorname{supp}(A\boxtimes B)=\operatorname{supp}(A)\times\operatorname{supp}(B). \tag{8.4.4}$$

For a geometric point (a,b), the compatibility of geometric pullback with derived tensor product gives

$$(A\boxtimes B)_{(a,b)}\xrightarrow{\sim}A_a\otimes^L_\Lambda B_b. \tag{8.4.5}$$

The right-hand side is zero if one of its factors is zero. Conversely, suppose that $A_a\neq 0$ and $B_b\neq 0$. If Λ is a field, then

$$H^n(A_a\otimes^L_\Lambda B_b)\simeq\bigoplus_{r+s=n}H^r(A_a)\otimes_\Lambda H^s(B_b). \tag{8.4.6}$$

Take r,s such that $H^r(A_a)\neq 0$ and $H^s(B_b)\neq 0$. The corresponding summand in degree $r+s$ is nonzero. Hence $A_a\otimes^L_\Lambda B_b\neq 0$.

Suppose now that Λ is finite local with residue field κ. The stalks A_a and B_b are perfect Λ-complexes. By derived Nakayama, we have

$$A_a\otimes^L_\Lambda\kappa\neq 0,\qquad B_b\otimes^L_\Lambda\kappa\neq 0.$$

By (8.4.6), their derived tensor product over κ is nonzero. By associativity,

$$\begin{aligned}&(A_a\otimes^L_\Lambda B_b)\otimes^L_\Lambda\kappa\\&\qquad\simeq(A_a\otimes^L_\Lambda\kappa)\otimes^L_\kappa(B_b\otimes^L_\Lambda\kappa)\neq 0.\end{aligned}$$

Therefore $A_a\otimes^L_\Lambda B_b\neq 0$. This proves (8.4.4).

Apply (8.4.4) to the fiberwise external product in (8.4.1). Since η_0 has one underlying topological point, projection from the η_0-factor gives

$$\begin{aligned}\mathrm{SS}_\mu(F\boxtimes G) &= \mathrm{pr}_{T^*(X\times Y)}\operatorname{supp}\big(\mu\mathcal{H}om(F,F)\boxtimes_{\eta_0}\mu\mathcal{H}om(G,G)\big)\\ &= \mathrm{pr}_{T^*X}\operatorname{supp}\mu\mathcal{H}om(F,F)\times\mathrm{pr}_{T^*Y}\operatorname{supp}\mu\mathcal{H}om(G,G).\end{aligned}$$

This is (8.4.2). The same argument applies to the field E.

It remains to prove (8.4.3). For the kernel $\mathscr{H}_F$, the identity of F, the canonical morphism $\Delta_X^! \to \Delta_X^*$, and evaluation give

$$\Lambda_X \xrightarrow{1_F} \Delta_X^!\mathscr{H}_F \longrightarrow \Delta_X^*\mathscr{H}_F \xrightarrow{\mathrm{ev}_F} K_{X/k}. \tag{8.4.7}$$

Define the analogous sequences for G and $F\boxtimes G$.

The compatibility of (8.2.1) with the identity morphisms gives a commutative diagram

$$\begin{tikzcd} \Lambda_X\boxtimes\Lambda_Y \arrow[r, "1_F\boxtimes 1_G"] \arrow[d, "\sim"'] & \Delta_X^!\mathscr{H}_F\boxtimes\Delta_Y^!\mathscr{H}_G \arrow[d, "\sim"] \\ \Lambda_{X\times Y} \arrow[r, "1_{F\boxtimes G}"'] & \Delta_{X\times Y}^!\mathscr{H}_{F\boxtimes G} \end{tikzcd} \tag{8.4.8}$$

The canonical morphisms $\Delta^! \to \Delta^*$ give a commutative diagram

$$\begin{tikzcd} \Delta_X^!\mathscr{H}_F\boxtimes\Delta_Y^!\mathscr{H}_G \arrow[r] \arrow[d, "\sim"'] & \Delta_X^*\mathscr{H}_F\boxtimes\Delta_Y^*\mathscr{H}_G \arrow[d, "\sim"] \\ \Delta_{X\times Y}^!\mathscr{H}_{F\boxtimes G} \arrow[r] & \Delta_{X\times Y}^*\mathscr{H}_{F\boxtimes G}. \end{tikzcd} \tag{8.4.9}$$

Its commutativity follows from the naturality of $\Delta^! \to \Delta^*$ and its compatibility with external products. The evaluation morphisms give a commutative diagram

$$\begin{tikzcd} \Delta_X^*\mathscr{H}_F\boxtimes\Delta_Y^*\mathscr{H}_G \arrow[r, "\mathrm{ev}_F\boxtimes\mathrm{ev}_G"] \arrow[d, "\sim"'] & K_{X/k}\boxtimes K_{Y/k} \arrow[d, "\sim"] \\ \Delta_{X\times Y}^*\mathscr{H}_{F\boxtimes G} \arrow[r, "\mathrm{ev}_{F\boxtimes G}"'] & K_{X\times Y/k}. \end{tikzcd} \tag{8.4.10}$$

The vertical morphisms are the external-product isomorphisms and (8.1.2). Diagram (8.4.8) follows from $\mathrm{id}_{F\boxtimes G} = \mathrm{id}_F\boxtimes\mathrm{id}_G$. To verify (8.4.10), use $\mathscr{H}_F \simeq \mathbb{D}_XF\boxtimes F$ and $\mathscr{H}_G \simeq \mathbb{D}_YG\boxtimes G$. Under $\mathbb{D}_{X\times Y}(F\boxtimes G)\simeq\mathbb{D}_XF\boxtimes\mathbb{D}_YG$, the evaluation of $F\boxtimes G$ is the composite of the symmetry isomorphism

$$(\mathbb{D}_XF\boxtimes\mathbb{D}_YG)\otimes(F\boxtimes G)\xrightarrow{\sim}(\mathbb{D}_XF\otimes F)\boxtimes(\mathbb{D}_YG\otimes G)$$

and $\mathrm{ev}_F\boxtimes\mathrm{ev}_G$, followed by (8.1.2). Hence (8.4.10) is commutative.

By adjunction, (8.4.7) defines a correspondence

$$\Delta_{X!}\Lambda_X \longrightarrow \mathscr{H}_F \longrightarrow \Delta_{X*}K_{X/k}. \tag{8.4.11}$$

Let $0_X : X \hookrightarrow TX$ be the zero section. Applying Verdier specialization to (8.4.11), we obtain

$$0_{X!}\Lambda_X \longrightarrow \nu_{\Delta_X}(\mathscr{H}_F) \longrightarrow 0_{X*}K_{X/k}. \tag{8.4.12}$$

Here we use the canonical isomorphisms $\nu_{\Delta_X}(\Delta_{X!}\Lambda_X)\simeq 0_{X!}\Lambda_X$ and $\nu_{\Delta_X}(\Delta_{X*}K_{X/k})\simeq 0_{X*}K_{X/k}$. Applying the unshifted Fourier transformation and the canonical isomorphisms

$$\mathfrak{F}_{\psi,TX/X}(0_{X!}\Lambda_X)\simeq\Lambda_{T^*X},\qquad \mathfrak{F}_{\psi,TX/X}(0_{X*}K_{X/k})\simeq e_X^{\vee*}K_{X/k} \tag{8.4.13}$$

we obtain morphisms

$$\Lambda_{T^*X} \xrightarrow{u_F} \mu\mathcal{H}om(F,F) \xrightarrow{v_F} e_X^{\vee*}K_{X/k}. \tag{8.4.14}$$

These are the two structural morphisms defining $\mathrm{CC}_\mu(F)$.

Put $\mathscr{M}_F = \mu\mathcal{H}om(F,F)$, $\mathscr{M}_G = \mu\mathcal{H}om(G,G)$, and $\mathscr{M}_{F\boxtimes G} = \mu\mathcal{H}om(F\boxtimes G, F\boxtimes G)$. Denote by

$$\Theta_{F,G} : \mathscr{M}_F \boxtimes_{\eta_0} \mathscr{M}_G \xrightarrow{\sim} \mathscr{M}_{F\boxtimes G}$$

the inverse of (8.4.1). The three commutative diagrams (8.4.8), (8.4.9), and (8.4.10) can be specialized simultaneously. In fact, the Künneth morphisms are the structural isomorphisms of the symmetric monoidal nearby-cycle functor [LZ22, Construction 3.3]. The naturality of the Fourier Künneth isomorphism in lemma 8.3 preserves the three resulting morphisms. Put

$$\mathscr{K}_F = e_X^{\vee *} K_{X/k}, \qquad \mathscr{K}_G = e_Y^{\vee *} K_{Y/k}, \qquad \mathscr{K}_{F\boxtimes G} = e_{X\times Y}^{\vee *} K_{X\times Y/k}.$$

We obtain a commutative diagram

$$\begin{tikzcd} \Lambda_{T^*X} \boxtimes \Lambda_{T^*Y} \arrow[r, "u_F\boxtimes u_G"] \arrow[d, "\sim"'] & \mathscr{M}_F \boxtimes \mathscr{M}_G \arrow[r, "v_F\boxtimes v_G"] \arrow[d, "\Theta_{F,G}"'] & \mathscr{K}_F \boxtimes \mathscr{K}_G \arrow[d, "\sim"] \\ \Lambda_{T^*(X\times Y)} \arrow[r, "u_{F\boxtimes G}"'] & \mathscr{M}_{F\boxtimes G} \arrow[r, "v_{F\boxtimes G}"'] & \mathscr{K}_{F\boxtimes G}. \end{tikzcd} \tag{8.4.15}$$

The factor η_0 is suppressed in the middle column. The left vertical morphism is induced by the external product of constant sheaves, and the right one is induced by (8.1.2). Thus (8.4.15) is compatible with both structural morphisms defining the microlocal trace class.

We now pass to cohomology with support. Put $C_F = \mathrm{SS}_\mu(F)$, $C_G = \mathrm{SS}_\mu(G)$, and let i_F, i_G be the corresponding closed immersions. By (8.4.2), the microlocal support of $F\boxtimes G$ is $C_F \times C_G$. By definition, $\mathrm{CC}_\mu(F)$ is represented on C_F by the composite

$$\Lambda_{C_F} \longrightarrow i_F^* \mathscr{M}_F \longrightarrow i_F^! e_X^{\vee *} K_{X/k}, \tag{8.4.16}$$

obtained from (8.4.14) by adjunction. By (8.4.15), the external product of (8.4.16) and its analogue for G is the corresponding composite on $C_F \times C_G$. Here we use the canonical isomorphism

$$\begin{aligned}(i_F^! e_X^{\vee *} K_{X/k}) &\boxtimes (i_G^! e_Y^{\vee *} K_{Y/k}) \\ &\xrightarrow{\sim} (i_F \times i_G)^! e_{X\times Y}^{\vee *} K_{X\times Y/k}.\end{aligned}$$

Taking H^0, the resulting class is $\mathrm{CC}_\mu(F) \boxtimes \mathrm{CC}_\mu(G)$. By the lower row of (8.4.15), it is also $\mathrm{CC}_\mu(F\boxtimes G)$. This proves (8.4.3).

For E-coefficients, take integral models as in the proof of lemma 8.2. The diagrams (8.4.8)–(8.4.15) commute modulo π^m for every m, compatibly with the transition maps. Derived completeness gives the corresponding integral diagrams, and inverting π gives them over E. The support equality follows from the stalk argument over the field E. Thus all three assertions hold with E-coefficients. □

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